NAAC grade ‘A+’

ACADEMIC REGULATIONS & SYLLABUS

Faculty of Science

P.D. Patel Institute of Applied Science

Department of Physical Sciences

Master of Science Program (Physics)

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Faculty of Science

ACADEMIC REGULATIONS

Master of Science (Physics) Programme

Charotar University of Science and Technology (CHARUSAT)
CHARUSAT Campus, At Post: Changa – 388421, Taluka: Petlad, District: Anand
Phone: 02697-247500, Fax: 02697-247100, Email: info@charusat.ac.in
www.charusat.ac.in

Year – 2023 Onwards

M. Sc. (Physics) Programme

CHARUSAT

FACULTY OF SCIENCE
ACADEMIC REGULATIONS

Faculty of Science

To ensure uniform system of education, duration of post graduate programmes, eligibility criteria for and mode of admission, credit load requirement and its distribution between course and system of examination and other related aspects, following are the academic rules and regulations.

1. *System of Education*

The Semester system of education shall be followed across The Charotar University of Science and Technology (CHARUSAT) at Master's levels. Each semester will be at least 90 working day duration. Every enrolled student will be required to do a specified course work in the chosen subject of specialization and also complete a project/dissertation if any. Medium of instruction will be English

2. *Duration of Programme*

Postgraduate programme	(M.Sc.)
Minimum	4 semesters (2 academic years)
Maximum	6 semesters (3 academic years)

The maximum limit can be extended by 1 or 2 semester subject to the approval of university on case to case basis.

3. *Eligibility for admissions*

For the admission to M.Sc., programs in the subject of Biological/Physical/Mathematical/Chemical Sciences a candidate must have obtained a Degree of Bachelor of Science from any recognized University or a Degree recognized as equivalent thereto, with minimum Second Class.

4. *Mode of admissions*

Admission to M.Sc. programme will be based on the performance at graduation.

5. Programme structure and Credits

A student admitted to a program should study the course and earn credits specified in the course structure.

6. Attendance

6.1 All activities prescribed under these regulations and listed by the course faculty members in their respective course outlines are compulsory for all students pursuing the courses. No exemption will be given to any student from attendance except on account of serious personal illness or accident or family calamity that may genuinely prevent a student from attending a particular session or a few sessions. However, such unexpected absence from classes and other activities will be required to be condoned by the Dean/Principal.

6.2 Student attendance in a course should be 80%.

7. Course Evaluation

7.1 The performance of every student in each course will be evaluated as follows:

7.1.1 Internal evaluation by the course faculty member(s) based on continuous assessment, for 50% of the marks for the course; and

7.1.2 Final examination will be conducted by the University for 70% of the marks for the course.

7.2 ***Internal Evaluation***

7.2.1 Internal evaluation will be based on internal tests and several other tools of assessment like, quiz, viva, seminar etc., as prescribed by concerned teacher and decided by the faculty.

7.3 ***Internal Institutional evaluation for practicals***

7.3.1 One internal practical test/viva will be conducted per semester totaling to 50 % internal marks for practicals

7.3.2 In "Continuous evaluation" Students shall be evaluated in a continuous manner for their involvement in the practical, aptitude for learning, completion of practical related assignments, regularity in the practicals and record keeping

7.4 ***University Examination***

7.3.3 The final examination by the University 50% of the evaluation for the course will be through written paper or practical test or oral test or presentation by the student or a combination of any two or more of these.

7.3.4 In order to earn the credit in a course a student has to obtain grade other than FF.

7.4 Performance at Internal & University Examination

7.4.1 Minimum performance with respect to internal marks as well as university examination will be an important consideration for passing a course. Details of minimum percentage of marks to be obtained in the examinations are as follows

Minimum marks in Internal Exam per subject	Minimum marks in University Exam per subject
36%	36%

7.5.2. If a candidate obtains minimum required marks per subject but fails to obtain minimum required overall marks, he/she has to repeat the university examination till the minimum required overall marks are obtained.

8 Grading

8.1 The internal evaluation marks and final University examination marks in each course will be converted to a letter grade on a ten-point scale as per the following scheme:

Grading Scheme:

Percentage	>=80	>=75 and <80	>=70 and <75	>=65 and <70	>=60 and <65	>=55 and <60	>=50 and <55	<50
Grade	**AA**	**AB**	**BB**	**BC**	**CC**	**CD**	**DD**	**FF**
Grade Point	10	9	8	7	6	5	4	0

For all CHARUSAT UG/PG degree/diploma programs, except Pharmacy programmes				**For all Pharmacy programmes**		
Letter Grade	**Grade Point**	**Grading Scheme for Mark (In %)**		**Letter Grade**	**Grade Point**	**Grading Scheme for Mark (In %)**
		For all CHARUSAT UG/PG degree/diploma programs, except Pharmacy and Nursing programmes	**B.Sc. Nursing, P.B. B.Sc. Nursing and M.Sc. Nursing**			B. Pharm, & M. Pharm.
O (Outstanding)	10	96.0-100	85.0 – 100	**AA** (Outstanding)	10	90.0 – 100
A+ (Excellent)	9	86.0-95.9	80.0 – 84.9	**AB** (Excellent)	9	80.0 – 89.9
A (Very Good)	8	76.0-85.9	75.0 – 79.9	**BB** (Very Good)	8	70.0 – 79.9
B+ (Good)	7	66.0-75.9	65.0 – 74.9	**BC** (Good)	7	60.0 – 69.9
B (Above Average)	6	56.0- 65.9	60.0 – 64.9	**CC** (Average)	6	50.0 – 59.9
C (Average)	5	46.0 – 55.9	50.0 – 59.9	**FF** (Fail)	0	Below 50.0
P (Pass)	4	36.0 – 45.9	-	**Ab** (Absent)	0	Absent
F (Fail)	0	Below 36.0	Below 50.0	-	-	-
Ab (Absent)	0	Absent	Absent	-	-	-

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY
Department of Physical Sciences
Faculty of Science
P. D. Patel Institute of Applied Sciences
MASTER OF SCIENCE

VISION

To become a pre-eminent national institute imparting science education integrated with research.

MISSION

To engage in education, research, and spread of science for the benefit of society.

PROGRAM EDUCATIONAL OBJECTIVES

The Postgraduate would be able to

PEO-1: Apply the acquired knowledge in various fields of sciences to plan and execute tasks.

PEO-2: Have scientific aptitude and competency in solving local and global problems.

PEO-3 Possess an ethical approach, while executing research projects and scientific writing.

PEO-4 Ability to follow life-long learning.

PEO-5 Understand the importance of team work and possess leadership and entrepreneurship traits.

PROGRAM OUTCOMES

The Post Graduates would be able to

PO1	**Knowledge in sciences:** Possess knowledge and comprehend the various core and allied courses offered under the opted stream of sciences.
PO2	**Planning Abilities:** Demonstrate effective planning abilities, including time management, resource management, delegation skills, organizational skills, teamwork, and interpersonal skills.
PO3	**Problem analysis:** Utilize the principles of scientific enquiry and think analytically, critically with clarity of thought, while solving problems.
PO4	**Modern tool usage:** **A**pply appropriate methodologies, methods and procedures, resources to conceptualize the matters.

PO5	**Leadership skills:** Realize the importance of teamwork and take initiatives to plan and implement a workflow to fulfill tasks with a compassionate attitude.
PO6	**Professional Identity:** Establish as a professional in the chosen area of the profession with social responsibility**.**
PO7	**Ethics in Science:** To understand the value of being ethical and implement its principles while carrying out research and publication, tasks at workplace and in life. To respect and adhere to the rules/regulations/treaties/directives/notifications/ laws related to Scientific research and business.
PO8	**Communication:** Communicate effectively and responsibly with the scientific community and with society at large, such as being able to comprehend and write effective reports, make effective presentations and documentation, and give and receive clear instructions.
PO9	**Modern Science and Society:** Apply reasoning informed by the contextual knowledge to assess societal, health, safety and legal issues and help society whenever situations arise.
PO10	**Environment and sustainability:** Understand the causes of environmental pollution and how it can be reduced and remedied using green technology in industries and environment.
PO11	**Life-long learning:** Vigilant about changes in the worldwide professional scenario arising out of technological, political, and economic factors. Recognize the need for and undertake steps necessary to fill the gaps recognized to keep oneself professionally competent. Self- assess and use feedback effectively from others to identify learning needs and to satisfy these needs on an ongoing basis.

PROGRAM SPECIFIC OUTCOMES (PSO)

PSO1: Ability to apply the fundamentals of physics through modelling, simulation and computing physical systems to understand their behaviour at different applied conditions.

PSO2: Ability to apply the skills developed during various courses on applied physics to understand and to provide viable solutions. Ability to tackle realistic problems by applying appropriate physical principles and scientific methodologies.

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

Department of Physical Sciences

Syllabus

M.Sc. Physics (Sem I-IV)

(Effective from July 2023)

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY (CHARUSAT)													
Subjects for Master of Science (Physics)													
Sem.	Subject	Subjects	Teaching scheme				Theory Evaluation			Practical Evaluation			Total I +II
			L+T	P	Contact hrs.	Credits	Institute	Univ.	Total-I	Institute	Univ.	Total-II	
I	PSUC501	Mathematical Methods of Physics	3+1	-	4	4	50	50	100	-	-	-	100
	PSUC502	Classical Mechanics	3+1	-	4	4	50	50	100	-	-	-	100
	PSUC503	Statistical Mechanics	3+1	-	4	4	50	50	100	-	-	-	100
	PSUC504	Electronics	3+1	-	4	4	50	50	100	-	-	-	100
	PSUS501	Physics Laboratory- 1	-	6	6	3	-	-	-	50	50	100	100
	PSUD501	Computer Programming	-	6	6	4	-	-	-	50	50	100	100
		Academic Speaking	-	2	2	2	-	-	-	50	50	100	100
		TOTAL			**30**	**25**			400			400	**700**

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY (CHARUSAT)													
Subjects for Master of Science (Physics)													
Sem	Subject	Subjects	Teaching scheme				Theory Evaluation			Practical Evaluation			Total I +II
			L+T	P	Contact hrs.	Credits	Institute	Univ.	Total-I	Institute	Univ.	To Contact hrs. tal-II	
II	PSUC551	Condensed Matter Physics	3+1	-	4	4	50	50	100	-	-	-	100
	PSUC552	Electrodynamics	3+1	-	4	4	50	50	100	-	-	-	100
	PSUC553	Quantum Mechanics I	3+1	-	4	4	50	50	100	-	-	-	100
	PSUA501	Analytical Techniques	4	-	4	4	50	50	100	-	-	-	100
	PSUD502	Computational physics	-	4	4	2	-	-	-	25	25	50	50
	PSUS502	Physics Laboratory- 2	-	8	8	4	-	-	-	50	50	100	100
	*****	Academic Writing	-	2	2	2	-	-	-	50	50	100	100
	######	University Elective	-	2	2	2	-	-	-				
		TOTAL			**32**	**26**			400				

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY (CHARUSAT)													
Subjects for Master of Science (Physics)													
Sem	Subject	Subjects	Teaching scheme				Theory Evaluation			Practical Evaluation			Total I +II
			L+T	P	Contact hrs.	Credits	Institute	Univ.	Total-I	Institute	Univ.	Total-II	
III	PSUC601	Nuclear and Particle Physics	3+1	-	4	4	50	50	100	-	-	-	100
	PSUC602	Atomic and Molecular Physics	3+1	-	4	4	50	50	100	-	-	-	100
	PSUC603	Quantum Mechanics II	3+1	-	4	4	50	50	100	-	-	-	100
	DE-1	Department Elective	3+1	-	4	4	50	50	100	-	-	-	100
	PSUS601	Physics Laboratory-3	-	6	6	3	-	-	-	50	50	100	100
	PSUP601	Minor Project	-	8	8	6	-	-	-	100	100	200	200
		TOTAL			**30**	**25**			400			300	**800**

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY (CHARUSAT)													
Subjects for Master of Science (Physics)													
Sem.	Subject	Subjects	Teaching scheme				Theory Evaluation			Practical Evaluation			Total I +II
			L+T	P	Contact hrs.	Credits	Institute	Univ.	Total-I	Institute	Univ.	Total-II	
IV	**DE-2	Departmental Elective	3+1	-	4	4	50	50	100	-	-	-	100
	**DE-3	Departmental Elective	3+1	-	4	4	50	50	100	-	-	-	100
	**DE-4	Departmental Elective	3+1	-	4	4	50	50	100	-	-	-	100
	**DE-5	Departmental Elective	3+1	-	4	4	50	50	100	-	-	-	100
	PSUS602	Physics Laboratory -4	-	14	14	9	-	-	-	200	200	400	400
		TOTAL			**30**	**25**							**600**

Subjects for Master of Science (Physics) -Research												
IV	PSUP602	Research Project	-	30	30	25	-	-	-	400	400	800	800
		TOTAL			**30**	**25**							**800**

General List of Departmental Electives (DE)	
PSE1	Thin film Technology
PSE2	Physics of Solar cells
PSE3	Nanomaterials: Synthesis, Properties and Applications
PSE4	Semiconductor Devices
PSE5	Experimental techniques
PSE6	Plasma Physics
PSE7	General Relativity, Black holes and Cosmology -I
PSE8	General Relativity, Black holes and Cosmology –II
PSE9	Vacuum Technology
CS291	Data Analytic
PSE52	Arduino Programming

SYLLABI
(Semester – 1)

PSUC501 Mathematical Methods of Physics

1.	Department/Centre/School proposing the course		PHYSICS
2.	Course Title		Mathematical Methods of Physics
3.	L-T-P structure		3-1-0
4.	Credits		4
5.	Course number		PSUC501
6.	Course Status (Course Category for Program) PG		
	Programme Core for:		M.Sc. Physics
7.	Pre-requisite(s)		NIL
8.	Status vis-à-vis other courses		
	8.1 List of courses precluded by taking this course (significant overlap)		
	(a)	Significant Overlap with any UG/PG course of the Dept./Centre/ School	NIL
	(b)	Significant Overlap with any UG/PG course of other Dept./Centre/ School	NIL
9.	Not allowed for		NIL
10.	Course offered in		Semester -1
11.	Faculty who will teach the course:		Dr. Krunal Kachhia/Dr. Tapobroto Bhanja/Dr. Shweta Dabhi
12.	Will the course require any visiting faculty?		No
13. *	Course contents: “Complex functions, Analytic function, Complex Integration, Cauchy’s theorem, residue theorem, Fourier Series, Fourier Transforms, Laplace Transforms, Tensors, Kronecker and Levi Civita tensors, inner and outer products, symmetric and antisymmetric tensors, covariant and contravariant tensors, group Theory, Fourier Gamma and Beta functions, Legendre, Hermite, Laguerre Polynomials, Binomial, Gaussian distributions, Application to the solutions of differential equations ”		
14.	Lecture Outline(with topics and number of lectures)		

	Module no.	Topic	No. of hours
	1	**Complex Analysis:** Complex functions, Analytic function, Complex Integration, Cauchy’s theorem, Laurent’s Theorem, Taylor's Theorem, Residue theorem, Harmonic Functions, transformations of elementary functions, Evaluation of definite integrals	10
	2	**Fourier Series and Integral Transforms**: Fourier series, Dirichlet Condition: convergence, Parseval’s identity, Fourier transforms, Laplace transforms, Applications of Integral transforms in physical situations.	12
	3	Homogeneous and Inhomogeneous differential equations and their solutions: partial Differential equations, Laplace and Poisson’s Equations, Green’s functions, Applications in physical phenomena, Gaussian Integrals, Legendre Polynomials, Bessel’s Functions, Hermite Polynomials, Laguerre Polynomials, Orthogonality and recurrence relations, Associated Legendre Polynomials , Applications	15
	4	**Probability distributions, Group Theory and Tensor**: Review of Probability theory - Binomial, Poisson and Normal Distributions. Basics of Groups theory, Isomorphism and homomorphism, group representation, Continuous Groups, Lorentz group, Lie Groups: U(1), SO(3), SU(2). Tensors - covariant and contravariant tensors, Kronecker and Levi Civita tensors, inner and outer products, symmetric and antisymmetric tensors, Metric tensor, Derivatives of tensors, Christofell’s symbols, Covariant derivatives, Geodesic equation.	20
15.	Brief description of tutorial activities:		

	Module no.	Description	No. of hours
	1	Analysis of Complex functions, Analytic functions, Cauchy theorem, Cauchy integral formula, Residue theorem.	04
	2	Fourier series and Integral Transforms	04
	3	Applications of Differential equations in Physics	05
	4	4 Vectors and tensors: Applications in Special Theory of Relativity	03
	5	Applications of Group Theory in Physics	03
16.	Brief description of Practical / Practice activities		
	Module no.	Description	No. of hours
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17.	Brief description of module-wise activities pertaining to self-learning components (Include topics that the students would do self-learning from books / resource materials: Do not Include assignments / term papers etc.)		
	Module no.	Description	
	Module 1	Real Functions, Complex Numbers	
	Module 2	Matrices	
18.	Suggested texts and reference materials STYLE: Author name and initials, Title, Edition, Publisher, Year		
	1. F. W. Byron and R. W. Fuller, Mathematics of Classical and Quantum Physics, Vol 1-2, Dover Publications, Inc., New York, 1992. 2. Complex Variables and Applications, J.W. Brown & R.V. Churchill, 7th Ed. 2003, Tata McGraw-Hill. 3. Mary L Boas , Mathematical Methods in Physical Sciences, 3rd Edition, John Wiley & Sons, India. 4. Arfken& Weber, Mathematical Methods for Physicists (6^{th} edition), Elsevier, Academic Press (2012). 5. Mathematical Methods for Physicists and Engineers, by K.F. Reily, M.P. Hobson and S.J. Bence.		
19.	Resources required for the course (itemized student access requirements, if any)		
	19.1	Software	C/C++/Fortran compilers, Codeblock
	19.2	Hardware	Computers
	19.3	Teaching aids (videos, etc.)	Not as such
	19.4	Laboratory	Computer Laboratory
	19.5	Equipment	Computers
	19.6	Classroom infrastructure	Black board and projector
	19.7	Site visits	Not required
	19.8	Others (please specify)	None
20.	Design content of the course (Percent of student time with examples, if possible)		
	Design-type problems	Nil	
	Open-ended problems	Nil	
	Project-type activity	Nil	
	Open-ended laboratory work	Nil	
	Others (please specify)	Nil	

Course Outcomes:
In this course the student will
CO1: learn about complex analysis and its applications in Physics
CO2: learn about vectors, vector spaces and Hilbert space and its applications in Quantum Mechanics
CO3: get introduced to various types of Tensors
CO4: learn about group theory and its applications in Physics
CO5: learn about Fourier transforms and their solving differential equations arise in Physics

Course Articulation Matrix											
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	-	3	3	-	3	3	3	3	3	3
CO2	3	-	3	3	-	3	3	3	3	3	-
CO3	3	-	3	3	-	-	3	3	3	-	3
CO4	3	-	3	3	-	-	3	3	3	3	3
CO5	3	-	3	3	-	3	3	3	3	3	3

PSUC502 CLASSICAL MECHANICS

1.	Department/Centre/School proposing the course		PHYSICS
2.	Course Title		Classical Mechanics
3.	L-T-P structure		3-1-0
4.	Credits		4
5.	Course number		PSUC502
6.	Course Status (Course Category for Program) PG		
	Programme Core for:		M.Sc. Physics
7.	Pre-requisite(s)		NIL
8.	Status vis-à-vis other courses		
	8.1 List of courses precluded by taking this course (significant overlap)		
	(a)	Significant Overlap with any UG/PG course of the Dept./Centre/ School	NIL
	(b)	Significant Overlap with any UG/PG course of other Dept./Centre/ School	NIL
9.	Not allowed for		NIL
10.	Course offered in		Semester 1
11.	Faculty who will teach the course: Dr. Kinnari Parekh, Dr. Tapobroto Bhanja, Dr. Manan Shah		
12.	Will the course require any visiting faculty?		No
13.	Course contents: Lagrangian and Hamiltonian Dynamics, Velocity Dependent Potentials, Symmetries and Noether's Theorem, Hamilton Jacobi Theory, Central Force Problems, Small Oscillations, Normal modes, Coupled and Anharmonic Oscillators, Rigid Body Dynamics, Special Theory of Relativity: Lorentz Transformations, Relativistic force and Einstein's Equation, Four Vectors (Covariant Formalism), Classical Chaos, Chaotic Trajectories and Lyapunov Exponents, Logistic Equations.		
14.	Lecture Outline(with topics and number of lectures)		

	Module no.	Topic	No. of hours
	1	**Lagrangian Mechanics**: Generalized coordinates constraints, principle of virtual work, Euler-Lagrange's equation, calculus of variations, Central force problem, Kepler problem: bound and scattering motions. Scattering in a central potential, Rutherford formula, scattering cross section. Lagrange's equation with undetermined multipliers, Velocity-dependent potentials, Noether's theorem and conserved currents.	10
	2	**Hamiltonian formulation:** Legendre transformations, Hamilton's equations, Symmetries and conservation laws in Hamiltonian picture, Hamilton's principle, phase space & phase space trajectories: Liouville's equation, Canonical transformations, Poisson brackets, Generating functions, Infinitesimal canonical transformations Conservation theorem in Poisson bracket formalism; Jacobi's identity, Applications to systems with one and two degrees of freedom. Focault's Pendulum	10

	3	**Hamilton-Jacobi Theory:** The Hamilton-Jacobi Equation for Hamilton's Principal Function, The Harmonic Oscillator Problem as an Example of the Hamilton-Jacobi Method, The Hamilton-Jacobi Equation for Hamilton's Characteristic Function, Separation of Variables in the Hamilton-Jacobi Equation Ignorable Coordinates and the Kepler Problem, Action-angle Variables in Systems of One Degree of Freedom, Action-Angle Variables for Completely Separable Systems, The Kepler Problem in Action-angle Variables. Transition from Classical to Quantum Mechanics.	12
	4	**Small Oscillations**: Basic concept of Small oscillations, Normal Modes and normal Frequencies, Coupled Oscillators, Forced Oscillations, Anharmonic oscillators.	10
	5	**Rigid body Dynamics**: Kinematics of rigid body motion: Euler Angles, Rotation matrices, Orthogonal transformations, Degrees of freedom of rigid body, Euler's theorem, Inertia tensor, Energy and Angular momentum, Coriolis Liouville's theorem.	10
	6	Lorentz Transformations, Relativistic kinematics. **Chaos:** Nonlinear dynamics, Logistic Equations, Chaotic motion, Lyapunov Exponents	8
	7		15
15.	Brief description of tutorial activities:		
	Module no.	Description	No. of hours
	1	Problem sessions constraints & D'Alembert's principle	1
	2	Problem sessions for Lagrangian equations of motion and its applications, Lagrange undetermined multipliers, symmetries and conservation laws	5
	3	Problems in Hamiltonian Dynamics	2
	4	Problems in Hamiltonian Jacobi Theory	2
	5	Problems in Small oscillations, normal modes, characteristic, rigid body dynamics	4
	6	Problems in Special Theory of Relativity	2
	7	Problems in classical chaos	3
16.	Brief description of Practical / Practice activities		
	Module no.	Description	No. of hours
	--	NIL	--
17.	Brief description of module-wise activities pertaining to self-learning component (Include topics that the students would do self learning from books / resource materials: Do not Include assignments / term papers etc.)		
	Module no.	Description	
18.	Suggested texts and reference materials STYLE: Author name and initials, Title, Edition, Publisher, Year		
	1. "Classical Mechanics" (Addison Wesley, Third Edition) - H. Goldstein, C. Poole and J. Safko. 2. "Mechanics (Theoretical Physics Vol. 1) - L. Landau and E. Lifschitz. 3. Classical Mechanics – (i) System of particles and Hamiltonian Dynamics and (ii) Point Particles and Relativity by Walter Greiner. 4. Classical Mechanics by G.Aruldhas, PHI 5. Classical Mechanics – N C Rana and P S Jog, Tata McGraw Hill, 1991 6. Classical Mechanics,- Yaswant Waghmere 7. Introductory Classical Mechanics with problems and solutions- David Morin 8. Classical Mechanics - Notling		

19.	Resources required for the course (itemized student access requirements, if any)		
	19.1	Software	NIL
	19.2	Hardware	NIL
	19.3	Teaching aids (videos, etc.)	NIL
	19.4	Laboratory	NIL
	19.5	Equipment	NIL
	19.6	Classroom infrastructure	Blackboard-choke
	19.7	Site visits	NIL
	19.8	Others (please specify)	NIL
20.	Design content of the course (Percent of student time with examples, if possible)		
	Design-type problems		NIL
	Open-ended problems		NIL
	Project-type activity		NIL
	Open-ended laboratory work		NIL
	Others (please specify)		NIL

Course Outcomes:

CO1: Learn the basic concepts of classical mechanics which includes constraints, degrees of freedom, generalized coordinates etc.

CO2: The student should be able to explain Lagrangian and Hamiltonian formulations, Poisson brackets and canonical transformations and to solve related problems.

CO3: The student will be able to develop an understanding of how to solve a given problem such as particle moving in space, motion of a body under central force fields, small oscillations, coupled oscillations etc.

CO4: Student will be able to describe rigid body dynamics and to solve given problems of rigid body motions.

Course Articulation Matrix											
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	-	-	3	3	3	3	-	-	-	3
CO2	3	-	3	3	3	-	3	-	-	-	3
CO3	3	-	3	3	-	3	-	-	-	-	3
CO4	3	-	3	3	-	-	-	3	-	-	3

PSUC503 Statistical Mechanics

1.	Department/Centre/School proposing the course		PHYSICS	
2.	Course Title		Statistical Mechanics	
3.	L-T-P structure		3-1-0	
4.	Credits		4	
5.	Course number		PS	
6.	Course Status (Course Category for Program) PG			
	Programme Core for:		M.Sc. Physics	
7.	Pre-requisite(s)		NIL	
8.	Status vis-à-vis other courses			
	8.1 List of courses precluded by taking this course (significant overlap)			
	(a)	Significant Overlap with any UG/PG course of the Dept./Centre/ School	NIL	
	(b)	Significant Overlap with any UG/PG course of other Dept./Centre/ School	NIL	
9.	Not allowed for		NIL	
10.	Course offered in		Semester-1	
11.	Faculty who will teach the course: Prof. P. C. Vinodkumar, Dr. Tapobroto Bhanja			
12.	Will the course require any visiting faculty?		No	
13.	Course contents: Microstates and macrostates, Canonical Ensembles, Gibbs's Paradox, Concepts of Phase Space and Liouville's theorem, elementary probability theory, distribution functions, central limit theorem, counting principles, fluctuations, statistics of paramagnetic systems, equipartition theorem, formulation of quantum statistics, introduction to density matrix, Bose-Einstein and Fermi-Dirac statistics, ideal Fermi gas, ideal Bose gas, Specific heats of solids, Fermi energy and mean energy, Fermi temperature, Electron Degeneracy pressure and white dwarfs , Phase Transitions, Virial Theorem, Ising Model, mean Field Theorem			
14.	Lecture Outline(with topics and number of lectures)			
	Module no.	Topic		No. of hours
	1	**Foundations of statistical Mechanics** – Microstates, macro states, phase space, statistical definition of entropy- Gibb's paradox, counting of microstates; phase space density, ergodic hypothesis, Liouville's theorem. Ensemble theory- microcanonical ensemble, entropy as an ensemble average, the uncertainty function, the canonical ensemble, grand canonical ensemble, system of non-interacting particles, calculation of observables as ensemble averages, Gibb's correction factor		13
	2	**Applications of classical statistical mechanics** – Fluctuations, Virial theorem and equipartition theorem, quantum systems in Boltzmann statistics, N-quantum mechanical harmonic oscillators, Paramagnetic behavior of substances, negative temperature in two-level systems, Gases with internal degrees of freedom-Translational, rotational and vibrational partition functions, relativistic ideal gas		15
	3	**Quantum Statistical Mechanics:** Density operators, properties of density matrix, Symmetry character of many particle systems. Ideal quantum systems: ideal Bose gas, ideal Fermi gas. Applications of Bose and Fermi gases.		13
	4	**Real gases and Phase Transitions:** Real gases, Mayer's Cluster expansion, Virial coefficients. Classification of Phase Transitions, examples of phase transitions, Critical indices. Ising model (one dimensional), Ising model in the mean field approximation, Heisenberg model in the mean field approximation, order-disorder phase transitions in the Ising model		15
15.	Brief description of tutorial activities:			
	Module no.	Description		No. of hours
	1	Fundamentals		2
	2	Harmonic Oscillators, Paramagnetic systems (classical)		2

	3	Problems involving Bose and Fermi gas	4
	4	Applications in Ising Model, mean field models	3
	5	Electron degeneracy and various applications	3
16.	Brief description of Practical / Practice activities		
	Module no.	Description	No. of hours
17.	Brief description of module-wise activities pertaining to self-learning components (Include topics that the students would do self-learning from books / resource materials: Do not Include assignments / term papers etc.)		
	Module no.	Description	
	Module 1	Basics of Probability, random walk, review of thermodynamics	
	Module 2, 3	Trace calculations, rotational and vibrational energy	
	Module 4	Virial Theorems, critical coefficients	
18.	Suggested texts and reference materials STYLE: Author name and initials, Title, Edition, Publisher, Year		
	1. R. K. Pathria and Paul D. Beale, Statistical Mechanics, 3rd Edition, Academic Press, 2012 2. F. Reif, Fundamentals of Statistical and Thermal Physics, International Student Ed., McGraw Hill. 3. Zemansky and Dittman, Heat and Thermodynamics 4. Sheng-Keng Ma, Statistical Mechanics, World Scientific		
19.	Resources required for the course (itemized student access requirements, if any)		
	19.1	Software	No software required
	19.2	Hardware	No hardware required
	19.3	Teaching aids (videos, etc.)	Not as such
	19.4	Laboratory	No Laboratory required
	19.5	Equipment	No equipment required
	19.6	Classroom infrastructure	No specific facility required
	19.7	Site visits	Not required
	19.8	Others (please specify)	None
20.	Design content of the course (Percent of student time with examples, if possible)		
	Design-type problems		Nil
	Open-ended problems		Nil
	Project-type activity		Nil

Course Outcomes:

CO1: The students should be able to understand counting using combinatorics principles

CO2: The students should be able to apply statistical mechanical tools to calculate thermodynamics of a given system

CO3: The students should be able to judge and apply appropriate statistics and ensemble to solve a particular problem

Course Articulation Matrix

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	-	3	-	-	-	-	-	-	-	-
CO2	3	-	3	-	-	-	-	-	-	-	-
CO3	3	-	3	-	-	-	-	-	-	-	-

PSUC504 Electronics

1.	Department/Centre/School proposing the course	PHYSICS	
2.	Course Title	Electronics	
3.	L-T-P structure	3-1-0	
4.	Credits	4	
5.	Course number	PSUC504	
6.	Course Status (Course Category for Program)	PG	
	Programme Core for:	M.Sc. Physics	
7.	Pre-requisite(s)	Knowledge of I-V characteristics of transistor and fundamental laws of electronics	
8.	Status vis-à-vis other courses		
	8.1 List of courses precluded by taking this course (significant overlap)		
	(a) Significant Overlap with any UG/PG course of the Dept./Centre/ School	Nil	
	(b) Significant Overlap with any UG/PG course of other Dept./Centre/ School	Nil	
9.	Not allowed for	--	
10.	Course offered in	Semester 1	
11.	Faculty who will teach the course: Dr. C K Sumesh, Dr. Manan Shah		
12.	Will the course require any visiting faculty?	No	
13.	Course contents: Linear and analog integrated circuits, timer circuits, frequency and phase modulations, digital circuits, shift registers, counters, introduction to microprocessor		
14.	Lecture Outline(with topics and number of lectures)		
	Module no.	Topic	No. of hours
	1	**Operational amplifier (IC-741) and its applications:** Basic operational amplifier, Operational amplifier with negative feedback, Voltage series feedback amplifier: close loop voltage gain, input resistance, output resistance, voltage follower. Voltage shunt feedback amplifier: close loop voltage gain, input resistance, output resistance, Inverter. Frequency compensation and slew rate, DC and AC amplifiers. Summing, Scaling and Averaging amplifier, Integrator and Differentiator circuit. Active filters: First order Low pass, High pass, and Band pass filters. Integrator and differentiator, Voltage to current and current to voltage converter, Schmitt trigger	18
	2	**Timer (IC-555) & its applications:** Internal Structure (Block Diagram) Operation, Astable, Monostable multivibrator and Applications.	05
	3	**Combinational digital circuit design:** Standard Gate Assemblies, Arithmetic Functions, Digital Comparator, Parity Checker -Generator, Multiplexer, De-multiplexer, Encoder, Decoder, Digital to Analog Converter, Analog to Digital Converter Tri-State buffer, Read Only Memory (ROM), ROM application	11
	4	**Sequential digital circuit design:** The clocked S-R flip-flop, D flip-flop, J-K flip-flop and T flip-flop, Shift Registers, Counters, Applications of counters	11
15.	Brief description of tutorial activities: Problem sessions and clarification of doubts		
	Module no.	Topic Description	No. of hours
	1	1. closed-loop non inverting op-amp 2. closed-loop inverting amplifier op-amp	32

		3. To design and verify a summing, and scaling amplifier circuit using inverting op-amp (IC741) configuration. 4. To design and verify an averaging amplifier circuit using inverting op-amp (IC741) configuration. 5. To design and verify a differentiator circuit using inverting op-amp (IC741) configuration and plot its output response for sine, square and triangular inputs. 6. To design and verify an integrator circuit using inverting op-amp (IC741) configuration and plot its output response for sine, square and triangular inputs. 7. To design and verify a high pass filter circuit using noninverting op-amp (IC741) configuration and plot its gain versus frequency graph. 8. To design and verify a low pass filter circuit using noninverting op-amp (IC741) configuration and plot its gain versus frequency graph.	
	2	1. To design and verify astable multivibrator circuit. 2. To design and verify monostable multivibrator circuit	04
	3	1. Study the characteristics of Half-adder and half-subtractor 2. Study the characteristics of full-adder and full-subtractor	08
	4	1. S-R Flip flop and D Flip flop 2. J-K Flip flop and T Flip flop 3. Shift-register 4. Counter	16
16.	Brief description of Practical / Practice activities: Refer to Laboratory Physics Course-1 : Analog and Digital Electronics		
	Module no.	Description	No. of hours
17.	Brief description of module-wise activities pertaining to self-learning component (Include topics that the students would do self-learning from books / resource materials: Do not Include assignments / term papers etc.)		
	Module no.	Description	
	1	Design aspect of experiments related to Op-Amp (IC-741)	
	2	Design aspect of experiments related to timer IC 555	
	3	Design aspect of experiments related to digital electronics	
	4	Design aspect of experiments related to digital electronics	
18.	Suggested texts and reference materials STYLE: Author name and initials, Title, Edition, Publisher, Year		
	1) Ramakant A.Gayakward- Op-Amps and linear integrated circuits-Pearson Education. 2) A. Anand Kumar - Fundamentals of Digital Circuits 3) Electronic Fundamentals and Applications : Integrated and Discrete Systems, John D. Ryder, PHI, Fifth edition. 4) Ramesh Goankar- Microprocessor Architecture, Programming and application with the 8085-Penram International Publishing Company.		
19.	Resources required for the course (itemized student access requirements, if any)		
	19.1	Software	–

	19.2	Hardware	--
	19.3	Teaching aids (videos, etc.)	yes
	19.4	Laboratory	Lab. Equipped to perform electronics experiments
	19.5	Equipment	Power supply, frequency generator, CRO, DMM, assorted items like bread board, wires, electronics components, ICs etc.
	19.6	Classroom infrastructure	No specific facility required
	19.7	Site visits	Not required
	19.8	Others (please specify)	None
20.	Design content of the course (Percent of student time with examples, if possible)		
	Design-type problems		25% of student time of tutorial/assignment: Basic circuit design exercise
	Open-ended problems		10-15% of student time
	Project-type activity		10-15% of student time
	Open-ended laboratory work		10-15% of student time
	Others (please specify)		Nil

Course Outcomes:

At the end of the course, the students will be able
CO1: to understand the fundamentals of working of op-amp and digital electronics
CO2: to explain various applications of analogy and digital electronic based devices
CO3: Design, development and verification of circuit, data measurements, data analysis, handling of laboratory equipments

Course Articulation Matrix											
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	3	3	3	-	3	-	3	3	-	3
CO2	3	3	3	3	-	3	-	3	3	-	3
CO3	3	3	3	3	3	3	3	3	3	-	-

PSUD501 Computer Programming

1.	Department/Centre/School proposing the course	PHYSICS	
2.	Course Title	Laboratory Physics Course 2: Computer Programming	
3.	L-T-P structure	1-1-4	
4.	Credits	4	
5.	Course number	PS729	
6.	Course Status (Course Category for Program) PG		
	Programme Core for:	M.Sc. Physics	
7.	Pre-requisite(s)	Preferable: Introduction to computer, types of programming languages	
8.	Status vis-à-vis other courses		
	8.1 List of courses precluded by taking this course (significant overlap)		
	(a)	Significant Overlap with any UG/PG course of the Dept./Centre/ School	NIL
	(b)	Significant Overlap with any UG/PG course of other Dept./Centre/ School	NIL
9.	Not allowed for	NIL	
10.	Course offered in	Semester 1	
11.	Faculty who will teach the course: Dr. Shweta Dabhi		
12.	Will the course require any visiting faculty?	No	
13.	Course contents: Introduction to language and simple programming, Control and loop structures, Arrays and functions		
14.	Lecture Outline(with topics and number of lectures)		
	Module no.	Topic	No. of hours
	1	Introduction to language and simple programming: Integer and floating point arithmetic, Precision, Variable types, Arithmetic statements, Input and output statements, Executable and non-executable statements, Operating systems	04
	2	Control and loop structures: Control structures	05
	3	Arrays and functions: Arrays, Repetitive and logical structures, Subroutines and functions, Operation with files	05
15.	Brief description of tutorial activities: Problem sessions and clarification of doubts.		
	Module no.	Description	No. of hours
	1	Introduction to language and simple programming:	04

		Integer and floating point arithmetic, Precision, Variable types, Arithmetic statements, Input and output statements,	
	2	Control and loop structures: Control structures, Executable and non-executable statements	05
	3	Arrays and functions: Arrays, Repetitive and logical structures, Subroutines and functions, Operation with files, Operating systems, Creation of executable programs	05
16.	Brief description of Practical / Practice activities		
	Module no.	Description	No. of hours
	1	Introduction to language and simple programming: Integer and floating point arithmetic, Precision, Variable types, Arithmetic statements, Input and output statements,	16
	2	Control and loop structures: Control structures, Executable and non-executable statements	20
	3	Arrays and functions: Arrays, Repetitive and logical structures, Subroutines and functions, Operation with files, Operating systems, Creation of executable programs	20
17.	Brief description of module-wise activities pertaining to self-learning component (Include topics that the students would do self-learning from books / resource materials: Do not Include assignments / term papers etc.)		
	Module no.	Description	
	1	Simple program including basic syntax	
	2	Programs including looping structure	
	3	Programs including concepts of arrays and functions	
18.	Suggested texts and reference materials :		
	1. Programming in C, E Balagurusamy, Tata McGraw-Hill Education 2. Let Us C, Yashwant P Kanetkar, BPB Publications 3. C Language and Numerical Methods by C. Xavier, New Age International Publication 4. Programming with C, Byron Gottfried, Schaum's outlines, McGraw Hill Publications 5. C for Beginners, Madhusudan Mothe, Shroff pubshers 6. Computer Oriented Numerical Methods, V. Rajaraman, PHI Learning Publisher		
19.	Resources required for the course (itemized student access requirements, if any)		
	19.1	Software	"C" or "C++" supported compiler
	19.2	Hardware	Computer system (Window 10 or higher)
	19.3	Teaching aids (videos, etc.)	Writing board, Projection System
	19.4	Laboratory	Computer lab
	19.5	Equipment	Nil

	19.6	Classroom infrastructure	Yes
	19.7	Site visits	Nil
	19.8	Others (please specify)	
20.	Design content of the course (Percent of student time with examples, if possible)		
	Design-type problems		20% of student time of practice hours
	Open-ended problems		Nil
	Project-type activity		Nil
	Open-ended laboratory work		Nil
	Others (please specify)		Nil

Course Outcomes:

At the end of the course, student would able to

CO1: implement the algorithms and draw flow charts for solving mathematical, physical and engineering problems

CO2: develop ability to design and develop computer programs

CO3: develop confidence for self-education and ability for life-long learning needed for computer languages.

Course Articulation Matrix											
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	-	-	-	-	-	-	-	-	-	-
CO2	-	3	-	3	-	3	-	-	3	-	-
CO3	-	-	-	-	-	3	-	3	3	-	3

Semester 2

PSUC551 Condensed Matter Physics

1.	Department/Centre/School proposing the course		PHYSICS
2.	Course Title		Solid State Physics
3.	L-T-P structure		3-1-0
4.	Credits		4
5.	Course number		PSUC551
6.	Course Status (Course Category for Program) PG		
	Programme Core for:		M.Sc. Physics
7.	Pre-requisite(s)		Module 0
8.	Status vis-à-vis other courses		
	8.1 List of courses precluded by taking this course (significant overlap)		
	(a)	Significant Overlap with any UG/PG course of the Dept./Centre/ School	NIL
	(b)	Significant Overlap with any UG/PG course of other Dept./Centre/ School	NIL
9.	Not allowed for		NIL
10.	Frequency of offering		Semester III
11.	Faculty who will teach the course: Dr. Shweta Dabhi, Dr. C K Sumesh, Dr. Vanaraj Solanki		
12.	Will the course require any visiting faculty?		No
13.	Course contents: Crystal Physics Revisit, Bonding of solids. Elastic properties, phonons, lattice specific heat. Free electron theory and electronic specific heat. Response and relaxation phenomena. Drude model of electrical and thermal conductivity. Hall effect and thermoelectric power. Electron motion in a periodic potential, band theory of solids: metals, insulators and semiconductors. Superconductivity: type-I and type-II superconductors. Josephson junctions. Superfluidity. Defects and dislocations. Ordered phases of matter: translational and orientational order, kinds of liquid crystalline order. Quasi crystals.		
14.	Lecture Outline(with topics and number of lectures)		
	Module no.	Topic	No. of hours
	1	Symmetry, Bravais lattices, Reciprocal lattice, Unit cell-Wigner -Seitz cell - Miller planes and spacing, characteristics of cubic cells, Interpretation of Bragg equation, Powder and single crystal diffraction methods – Diffractometers	08
	2	Lattice Vibrations, quantization, phonon, Thermal Properties, Elastic properties, lattice specific heat, Drude model of electrical and thermal conductivity, electronic specific heat, Response and relaxation phenomena.	08

	3	Free Electron theory, Dependence of electron energy on the wave vector, E-K diagram. Free electron theory of metals, Thermal and Electrical transport properties, Electronic specific heat, Fermi surface. Hall effect. Thermionic emission. Failures of free electron theory, Energy Band Theory: Electron motion in a periodic potential, Bloch theorem, Kronig-Penny Model, band theory of solids: metals, insulators and semiconductors. Effective mass, Fermi surface and Density of states	10
	4	Terms and definitions used in magnetism, classification of magnetic materials, Atomic theory of magnetism– Langevin's classical theory of diamagnetism, Langevin's classical theory of paramagnetism, Quantum theory of paramagnetism, Ferromagnetism, Weiss molecular field,Temperature dependence of spontaneous magnetization, Ferromagnetic domains , Domain theory, Antiferromagnetism , Ferrimagnetism ,Structure of ferrite	10
	5	Occurrence of superconductivity, Meissner effect, Type I and Type II superconductors, Thermodynamics of the superconducting transition, London equation, Coherence length, BCS theory of superconductivity, Josephson effect, Macroscopic quantum interference – High temperature superconductors, Applications	10
		Total hours	60
15.	Brief description of tutorial activities:		
	Module no.	Description	No. of hours
		Total Tutorial hours (14 times 'T')	
16.	Brief description of Practical / Practice activities		
	Module no.	Description	No. of hours
17.	Brief description of module-wise activities pertaining to self-learning component :		
	Module no.	Description	
18.	Suggested texts and reference materials STYLE: Author name and initials, Title, Edition, Publisher, Year		
	1. Kittel, C.: Introduction to Solid State Physics, Wiley (2007). 2. Ashcroft and Mermin: Solid state Physics, Thomson (2007). 3. Ali Omar: Elementary Solid State Physics, Addison-Wesley (2005) 4. Elements of Solid State Physics, J.P. Srivastava, Prentice Hall of India (2nd Edition) 5. Solid state physics, A J Dekkar, Prentice Hall of India		
19.	Resources required for the course (itemized student access requirements, if any)		
	19.1	Software	-
	19.2	Hardware	-
	19.3	Teaching aids (videos, etc.)	Crystal structure, NPTEL vidoes
	19.4	Laboratory	-

	19.5	Equipment	Type of equipment required, number of access points, etc.
	19.6	Classroom infrastructure	Blackboard-choke
	19.7	Site visits	National Material research Laboratory
	19.8	Others (please specify)	
20.	Design content of the course (Percent of student time with examples, if possible)		
	Design-type problems		-
	Open-ended problems		
	Project-type activity		
	Open-ended laboratory work		
	Others (please specify)		

Course Outcomes:

CO1: Student will acquire sound knowledge in analyzing crystal structure data for some basic crystals using X-ray.

CO2: Student should be able to understand importance of quantum mechanical treatment to understand solid state physics.

CO3: Basics origin of superconductivity and how to tune the properties.

CO4: How electrical and thermal properties of materials change at nanoscale.

CO5: At the end students would gain experience in the problem solution which will help him to answer few JRF examination questions related to solid state physics.

Course Articulation Matrix

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	-	-	-	-	3	-	-	-	-	3
CO2	-	-	3	-	-	3	-	-	-	-	3
CO3	3	-	3	-	-	3	-	-	-	3	3
CO4	3	-	3	-	-	3	-	-	-	3	3
CO5	-	-	3	-	-	3	-	-	-	-	3

PSUC552 Electrodynamics

1.	Department/Centre/School proposing the course	PHYSICS
2.	Course Title	Electrodynamics
3.	L-T-P structure	3-1-0
4.	Credits	4
5.	Course number	PS
6.	Course Status (Course Category for Program) PG	
	Programme Core for:	M.Sc. Physics
7.	Pre-requisite(s)	NIL
8.	S tatus vis-à-vis other courses	
	8.1 List of courses precluded by taking this course (significant overlap)	
	(a) Significant Overlap with any UG/PG course of the Dept./Centre/ School	NIL
	(b) Significant Overlap with any UG/PG course of other Dept./Centre/ School	NIL
9.	Not allowed for	NIL
10.	Course offered in	Semester 2
11.	Faculty who will teach the course: Dr. Manan Shah/Dr. Tapobroto Bhanja	
12.	Will the course require any visiting faculty?	No
13.	Course contents: Review of electrostatics and Magnetostatics, Faraday's laws, Maxwell's equations, Scalar and vector potentials, Concepts of Gauge and Gauge invariance, Poynting's theorem and conservation laws, boundary conditions, electromagnetic wave propagation in space and medium, reflection and transmission of EM waves on an interface, wave guides, resonant cavities, Lorentz transformations, covariance of electrodynamics: 4-vectors and tensors, covariant formulation of Electrodynamics, Lagrangian for EM field, motion of a charged particle in EM fields, electric dipole, electric quadrupole and magnetic dipole radiation, radiation by a moving charge	
14.	Lecture Outline(with topics and number of lectures)	

	Module no.	Topic	No. of hours
	1	Maxwell's equations, Poynting theorem, Momentum conservation, Maxwell's stress tensor – Angular momentum. Electromagnetic Waves in Vacuum: Wave equation for E and B fields, Monochromatic plane waves, Energy & momentum in electromagnetic waves, Polarization of electromagnetic waves: linear and circular polarization. Electromagnetic waves in matter (Dielectric and conductor), Propagation in linear media-Boundary conditions, Reflection and Refraction – Snell's law. Reflection and Transmission at normal and oblique incidence – Fresnel's equations.	12
	2	Total internal reflections, EM waves in isotropic linear conducting media – Reflection at conducting surface. WaveGuides and Resonant Cavities: Bounded waves – TE, TM, TEM modes, Rectangular waveguides, Circular cylindrical waveguides, Resonant cavities, Q of a cavity, Dielectric waveguides (Optical fiber), H E Modes.	10
	3	Electromagnetic Radiation: Electromagnetic potentials Scalar and vector potentials, Covariant form of Maxwell's equations Gauge transformations, Gauge conditions (Lorentz and Coulomb gauges). Retarded Potentials, Jefimenko's equations for the E and B fields. Radiations from extended sources: Electric dipole radiation Magnetic dipole and electric quadrupole	13

		radiations, Center-Fed linear antenna, Hertzian dipole antenna, Small loop antenna.	
	4	Radiation from Moving point charges: Lienard-Wiechert potentials, Fields of a moving point charge, Power radiated by a point charge (Larmor formula), Radiation from a slowly moving charges, Radiation from relativistically moving charges, Larmor's generalization to relativistic case, Synchrotron radiation, Bremsstrahlung radiation. Scattering of radiation from quasi free charges (Thomson Scattering), Cerenkov radiation. Radiation reaction: Abraham –Lorentz formula, Physical basis of radiation reaction.	15
15.	Brief description of tutorial activities:		
	Module no.	Description	No. of hours
	1	Tutorials will be discussed on electrostatics and magnetostatics and other topics	3
	2	Discussions of application of various concepts of different topics to solve numerical problems	2
	3	Problem solution sessions and clarification of doubts	2
	4	Problem solution sessions and clarification of doubts	3
16.	Brief description of Practical / Practice activities		
	Module no.	Description	No. of hours
	Module 1	Determine the Velocity of EM waves for the given Co-axial Cable.	4
	Module 2	(a) To study the characteristics of the reflex klystron tube and to determine its electronic tuning range. (b) To study variations in its attenuation with the frequency. (c) To determine the frequency and wavelength of a microwave in a rectangular waveguide operated in TE_{10} mode.	10
	Module 3	Arranging the setup and performing the functional checks of Antenna system.	8
	Module 4	(a) To verify the law of Malus for plane polarized light. (b) To analyze elliptically polarized Light by using a Babinet's compensator. (c) To study the nature of polarization of laser light using photocell and quarter wave plate.	8

17.	Brief description of module-wise activities pertaining to self-learning component (Include topics that the students would do self-learning from books / resource materials: Do not Include assignments / term papers etc.)	
	Module no.	Description
	Module 1	Electrostatics and Magnetostatics
18.	Suggested texts and reference materials STYLE: Author name and initials, Title, Edition, Publisher, Year	
	1. David J. Griffiths, Introduction to Electrodynamics, 4th edition, Pearson New International Edition, 2014 2. Matthew N. O. Sadiku, Principles of Electromagnetics, 4th edition, Oxford international Students, 2009 3. Walter Greiner, Classical electrodynamics, Springer, Paperback, 2006	

19.	Resources required for the course (itemized student access requirements, if any)		
	19.1	Software	No software required
	19.2	Hardware	No hardware required
	19.3	Teaching aids (videos, etc.)	Not as such
	19.4	Laboratory	No Laboratory required
	19.5	Equipment	No equipment required
	19.6	Classroom infrastructure	No specific facility required
	19.7	Site visits	Not required
	19.8	Others (please specify)	None

20.	Design content of the course (Percent of student time with examples, if possible)	
	Design-type problems	Nil
	Open-ended problems	Nil
	Project-type activity	Nil
	¤ Open-ended laboratory work	Nil
	Others (please specify)	Nil

Course Outcomes:

CO1: The students will have the knowledge and skills to have an advanced understanding of Maxwell's equations.

CO2: The student would be able to understand the propagation of electromagnetic waves in materials, boundary conditions at material interfaces and the concept of waveguides.

CO3: The students will have learned covariant (relativistic formulation) of electrodynamics including four vectors, the electrodynamics field tensor, Lagrangian for EM fields, Lorentz transformations.

CO4: The student should be able to explain electric dipole and quadrupole radiation, magnetic dipole radiation and radiation by moving charge.

Course Articulation Matrix											
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	-	-	3	-	3	-	-	3	-	3
CO2	3	-	3	-	-	3	-	-	3	-	3
CO3	3	-	-	3	-	3	-	-	3	-	3
CO4	3	-	-	3	-	3	-	-	3	-	3

PSUC553 Quantum Mechanics – I

1.	Department/Centre/School proposing the course		PHYSICS
2.	Course Title		Quantum Mechanics – I
3.	L-T-P structure		3-1-0
4.	Credits		4
5.	Course number		PSUC553
6.	Course Status (Course Category for Program) PG		
	Programme Core for:		M.Sc. Physics
7.	Pre-requisite(s)		Concepts of vector space
8.	Status vis-à-vis other courses		
	8.1 List of courses precluded by taking this course (significant overlap)		
	(a)	Significant Overlap with any UG/PG course of the Dept./Centre/ School	NIL
	(b)	Significant Overlap with any UG/PG course of other Dept./Centre/ School	NIL
9.	Not allowed for		NIL
10.	Course offered in		Semester -2
11.	Faculty who will teach the course: Prof. P. C. Vinodkumar/Dr. Tapobroto Bhanja/ Dr. Shweta Dabhi		
12.	Will the course require any visiting faculty?		No
13.	Course contents: Postulates of quantum mechanics, one-dimensional problems, Tunneling problem, 3D problems, symmetry and degeneracy, Angular momentum algebra, Clebsch Gordon Coefficients, Hydrogen atom, Time independent and time dependent perturbation theory, Zeeman and WKB Approximation, Variational method		
14.	Lecture Outline(with topics and number of lectures)		
	Module no.	Topic	No. of hours
	1	Hilbert space, Dirac notations and bra ket algebra	2
	2	Postulates of quantum mechanics, one dimensional problems	6
	3	3D problems, symmetry and degeneracy, Quantum Tunneling	8
	4	Angular momentum algebra, Hydrogen atom	8
	5	Zeeman Effect, WKB Formalism, Variational Method	6
	6	Time Independent Perturbation	10
15.	Brief description of tutorial activities:		
	Module no.	Description	No. of hours
	1	Tutorials and problem solution related to eigenvalue problem, linear independency and linear dependency of vectors, basis transformation	2
	2	Problem sessions and clarification of doubts related to the postulates of quantum mechanics and their applications	2
	3	Discussion of various 3D problems	2
	4	Tutorials on angular momentum algebra and related numerical of hydrogenic systems	2
	5	Problem sessions and clarification of doubts.	2
	6	Problem sessions and clarification of doubts.	2
16.	Brief description of Practical / Practice activities		
	Module no.	Description	No. of hours
	Module 2	Solving Schrodinger equation and plotting wave functions and probability distributions using scilab for some simple problems.	2

	Module 4	Plotting wave functions and probabilities of hydrogen atom energy states using scilab/mathematica.	2
17.	Brief description of module-wise activities pertaining to self-learning component (Include topics that the students would do self-learning from books / resource materials: Do not Include assignments / term papers etc.)		
	Module no.	Description	
	Module 1	Matrix algebra: symmetric and antisymmetric matrix, Hermitian matrix, unitary matrix, identity matrix, inverse of a matrix, solving eigenvalue problem, diagonalization of a matrix	
	Module 2	Properties of wave functions and implication of boundary conditions in solving differential equations	
	Module 3	Coordinate transformations and derivation of the angular momentum operators in terms of spherical coordinates	
18.	Suggested texts and reference materials STYLE: Author name and initials, Title, Edition, Publisher, Year		
	1. *Quantum Mechanics Concepts and Applications, N.* Zettilli, (Wiley) 2. Introductory Quantum Mechanics, Richard L. Liboff (Pearson). 3. *Quantum Mechanics, David J. Griffiths (Cambridge University)* 4. *Quantum Mechanics*, L. I. Schiff, (McGraw-Hill).		
19.	Resources required for the course (itemized student access requirements, if any)		
	19.1	Software	Scilab/mathematics
	19.2	Hardware	No hardware required
	19.3	Teaching aids (videos, etc.)	Not as such
	19.4	Laboratory	Computer Laboratory
	19.5	Equipment	Computers
	19.6	Classroom infrastructure	No specific facility required
	19.7	Site visits	Not required
	19.8	Others (please specify)	None
20.	Design content of the course (Percent of student time with examples, if possible)		
	Design-type problems	Nil	
	Open-ended problems	Nil	
	Project-type activity	Nil	

Course Outcomes:

CO1: At the end of the course, student will be able to understand the basic concepts of quantum mechanics which includes wavefunction and probability, postulates in quantum Mechanics, operators, eigen value equation etc.

CO2: Solve basic problems in quantum mechanics such as particle in a box, harmonic oscillator hydrogen atom, hydrogen atom in electric fields etc.

CO3: Student will learn advance quantum mechanical theory like variational method, perturbation method etc.

Course Articulation Matrix											
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	-	-	-	-	3	-	3	-	-	3
CO2	3	-	3	-	-	3	-	-	-	-	3
CO3	3	-	3	-	-	3	-	-	-	-	3

PSUA501 Analytical Techniques

1.	Department/Centre/School proposing the course	PHYSICS
2.	Course Title	Analytical Techniques lab
3.	L-T-P structure	4-0-0
4.	Credits	4
5.	Course number	PSUA501
6.	Course Status (Course Category for Program) PG	
	Programme Core for:	M.Sc. Physics
7.	Pre-requisite(s)	Theoretical background of various techniques
8.	Status vis-à-vis other courses	
	8.1 List of courses precluded by taking this course (significant overlap)	
	(a) Significant Overlap with any UG/PG course of the Dept./Centre/ School	NIL
	(b) Significant Overlap with any UG/PG course of other Dept./Centre/ School	NIL
9.	Not allowed for	NIL
10.	Course offered in	Elective
11.	Faculty who will teach the course:	
12.	Will the course require any visiting faculty?	No
13.	Course contents: Determination of various properties of materials using various analytical techniques such as UV-VIS, XRD, FTIR, Particle size analyzer, SEM, and VSM	
14.	Lecture Outline with topics	

Module no.	Description	No. of hours
1.	**UV-Vis Spectrophotometer** • Understanding operation of UV-VIS spectrophotometer, sample preparation, and standard operating procedure • Introduction for measurement of absorbance, reflectance and transmittance using UV-Vis spectrophotometer Band gap determination of thin film sample	7
2.	**X-ray Diffractometer (XRD)** • Understanding operation of XRD, sample preparation, and standard operating procedure • Crystal structure identification (cubic, hexagonal, triclinic, etc) of prepared powder sample using XRD sample	7
3.	**Particle size analyzer: (Dynamic light scattering)** • Understanding operation of Particle size analyzer, sample preparation, and standard operating procedure • Measuring particle size of prepared solution by Particle size analyzer	7

	4.	**Fourier transform infrared spectroscopy (FTIR)** • Understanding operation of FTIR, sample preparation, and standard operating procedure • Determination of composition in compound using molecular bond identification • Sample validation, identification of presence of elements in prepared sample	7
	5.	**Scanning Electron Microscope (SEM)** • Understanding operation of SEM, sample preparation, and standard operating procedure • Measuring morphological properties of powder sample by SEM • Measuring morphological properties of thin film (top view and cross-sectional view) sample by SEM	6
	6.	**Vibrating-sample Magnetometer (VSM)** • Understanding operation of VSM, sample preparation, and standard operating procedure • Measuring magnetic properties of prepared sample by VSM	6
16.	Suggested texts and reference materials :		
	1. Handbook of Analytical Instruments, R S Khandpur, McGraw Hill Education (India) Private Limited 2. X-ray Diffraction, A practical Approach by C Suryanarayana & M Grant Norton, Plenum Press, New York, 1998. 3. Introduction to magnetic materials by B D Cullity & C D Graham, Addision Wesley 4. Introduction to Instrumental Analysis by R. D. Broun, Mc Graw Hill (1987) 5. Principles of Instrumental Analysis, Skkog, Holler, Nieman, (Sixth Ed.) 6. Quantitative Inorganic Analysis including Elementary Instrumental analysis, By A. I. Vogel, 3ed, ELBS, 1964. 7. Instrumental methods of analysis, R. D. Braun 8. Analytical spectroscopy by Kamalesh Bansal- First edition 9. Electron microscopy in the study of material, P. J Grundy and G. A Jones, Edward Arnold 10. Spectroscopic identification of organic compounds Fifth Ed., Silvestrine, Bassler, Morrill, John Wiley and sons. 11. Analytical Chemistry by Gary Christain, 6th edition, 2008		
17.	Resources required for the course (itemized student access requirements, if any)		
	17.1	Software	Instrument wise different software
	17.2	Hardware	Computer system
	17.3	Teaching aids (videos, etc.)	Writing board, Projection System
	17.4	Laboratory	Sophisticated research instruments
	17.5	Equipment	Nil
	17.6	Classroom infrastructure	Yes
	17.7	Site visits	Nil
	17.8	Others (please specify)	

18.	Design content of the course (Percent of student time with examples, if possible)	
	Design-type problems	20% of student time of practice hours
	Open-ended problems	Nil
	Project-type activity	Nil
	Open-ended laboratory work	40% of student time of practice hours
	Others (please specify)	Nil

Course Outcomes:
At the end of the course, student would able to
CO1: Apply theoretical principles to analysis and measurements performed in the research laboratory.
CO2: Develop analytical skills for interpreting data and drawing inferences through team work.
CO3: Apply various procedures and techniques for the Hands-on experimental training of sophisticated instruments.

Course Articulation Matrix											
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	3	3	3	-	-	-	3	3	-	-
CO2	3	3	3	3	-	-	3	3	3	-	-
CO3	3	3	3	3	-	-	-	3	3	-	-

PSUD502 Computational physics

21.	Department/Centre/School proposing the course		PHYSICS
22.	Course Title		Computational Physics
23.	L-T-P structure		0-0-4
24.	Credits		2
25.	Course number		PSUD502
26.	Course Status (Course Category for Program) PG		
	Programme Core for:		M.Sc. Physics
27.	Pre-requisite(s)		Preferable: Introduction to computer,Matlab or Scilab basics
28.	Status vis-à-vis other courses		
	28.1 List of courses precluded by taking this course (significant overlap)		
	(a)	Significant Overlap with any UG/PG course of the Dept./Centre/ School	NIL
	(b)	Significant Overlap with any UG/PG course of other Dept./Centre/ School	NIL
29.	Not allowed for		NIL
30.	Course offered in		Semester 1
31.	Faculty who will teach the course: Dr. Shweta Dabhi, Dr. Tapobroto Bhanja, Dr. Manan Shah		
32.	Will the course require any visiting faculty?		No
33.	Course contents:		
34.	Lecture Outline(with topics and number of lectures)		

	Module no.	Topic	No. of hours
	1		
35.	Brief description of tutorial activities: Problem sessions and clarification of doubts.		
	Module no.	Description	No. of hours
	1		
36.	Brief description of Practical / Practice activities		
	Module no.	Description	No. of hours
	1	Programs related classical and quantum mechanics	20
	2	Programs related to statistical mechanics	20
	3	Programs related to advanced quantum mechanics	20
37.	Brief description of module-wise activities pertaining to self-learning component (Include topics that the students would do self-learning from books / resource materials: Do not Include assignments / term papers etc.)		
	Module no.	Description	

	1	
38.	Suggested texts and reference materials :	
	7. Computer Oriented Numerical Methods, V. Rajaraman, PHI Learning Publisher	
39.	Resources required for the course (itemized student access requirements, if any)	

	19.9	Software	Matlab and Scilab
	19.10	Hardware	Computer system Ubuntu/Window 10 or higher
	19.11	Teaching aids (videos, etc.)	Writing board, Projection System
	19.12	Laboratory	Computer lab
	19.13	Equipment	Nil
	19.14	Classroom infrastructure	Yes
	19.15	Site visits	Nil
	19.16	Others (please specify)	

40.	Design content of the course (Percent of student time with examples, if possible)	
	Design-type problems	20% of student time of practice hours
	Open-ended problems	Nil
	Project-type activity	Nil
	Open-ended laboratory work	Nil
	Others (please specify)	Nil

Course Outcomes:

At the end of the course, student would able to

CO1: **Use** computational tools for solving mathematical, physical and engineering problems

CO2: **Develop** problem solving skills and ability to design and develop computer programs

CO3: **Develop** confidence for self-education and ability for life-long learning needed for computer languages.

Course Articulation Matrix

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	-	-	-	-	-	-	-	-	-	-
CO2	-	3	-	3	-	3	-	-	3	-	-
CO3	-	-	-	-	-	3	-	3	3	-	3

Semester 3

PSUC601 NUCLEAR AND PARTICLE PHYSICS

1.	Department/Centre/School proposing the course		PHYSICS
2.	Course Title		NUCLEAR AND PARTICLE PHYSICS
3.	L-T-P structure		3-1-0
4.	Credits		4
5.	Course number		PS822
6.	Course Status (Course Category for Program) PG		
	Programme Core for:		M.Sc. Physics
7.	Pre-requisite(s)		NIL
8.	Status vis-à-vis other courses		
	8.1 List of courses precluded by taking this course (significant overlap)		
	(a)	Significant Overlap with any UG/PG course of the Dept./Centre/ School	NIL
	(b)	Significant Overlap with any UG/PG course of other Dept./Centre/ School	NIL
9.	Not allowed for		NIL
10.	Frequency of offering		Semester III
11.	Faculty who will teach the course: Dr. Manan Shah, Prof. P C Vinodkumar		
12.	Will the course require any visiting faculty?		No
13.	Course contents: Review of scattering theory and relativistic quantum mechanics, basic nuclear properties, nuclear force, nuclear models, radioactivity, nuclear reactions and nuclear decays, nuclear fission, nuclear fusion,nuclear detectors and accelerators; Particle physics: Basic interactions, classification of particles, Symmetries and conservation laws, Quark model for hadrons, basics of the standard mopdel of particle physics		
14.	Lecture Outline(with topics and number of lectures)		

	Module no.	Topic	No. of hours
	1	**Basic nuclear properties, two body problems in Nuclear Physics** 1.1 Basic nuclear properties 1.2 Experimental status of deuteron 1.3 The Schrödinger wave equation and its solution for deuteron 1.4 shape of the ground state wave function and Normalization of deuteron wave function 1.5 The radius and properties of deuteron, Mixing of orbitals in deuteron 1.6 Magnetic moment of Deuteron, Quadrupole moment of deuteron.	06

	2	**Nuclear force, nuclear models** 2.1 charge independence, charge symmetry, Non central (Tensor) force, Exchange forces 2.2 Nuclear models: Liquid drop model (review), Shell model, Explanation of nuclear data, Nordheim's rules for odd Z- odd N nuclei, islands of isomerism, Successes and failures of shell model. 2.2.1 Collective model, Rotational motion of the nucleus, Vibration of spherical nuclei, classification of vibration of spherical nuclei.	08
	3	**Basic Nuclear decays and nuclear reaction** 3.1 Basic alpha decay processes alpha decay systematic, GeigerNuttall Law, Theory of alpha decay, 3.2 Experimental information about beta decay, Energy released in beta decay (Q-value), Continuous beta spectrum and neutrino hypothesis, Fermi theory of beta-decay, Fermi-Kurie plot,, Experimental detection of neutrino 3.3 Gamma decay: Electromagnetic transitions, Radiation field multipolarity, 3.4 selection rules 3.5 Direct reaction, pick up reaction, Transfer and knock out reactions Fission - Fusion Reactions, compound nucleus	14
	4	**Nuclear detectors and Accelerators** 4.1 Self-study topics: Geiger Muller Counter(Gas-filled counters) , Scintillation detectors, Solid state X-ray and gamma-ray detectors, Neutron detectors, Neutrino Detectors, High-energy charged particle detectors, charge particle motion in matter 4.2 Van de Graff, Linear accelerator, Cyclotrons, High energy accelerators, Proton Synchrotron & LHC	08
	5	**Symmetries with the Quark model and standard model** 5.1 Classification of fundamental forces, Elementary particles and their quantum numbers, Conservation laws, Gellmann-Nishijima formula, particle symmetries 5.2 Quark model, conservation laws	14
		Total hours	50
15.	Brief description of tutorial activities:		

	Module no.	Description	No. of hours
	1	Tutorials on basic nuclear properties	2
	2	Conceptual question discussions	2
	3	Problem solution on radioactive decay	2
	4	Detectors and accelerator	2
	5	Question answer sessions on particle physics (Collider)	2
	Total Tutorial hours (14 times ‘T’)		10
16.	Brief description of Practical / Practice activities		
	Module no.	Description	No. of hours
	Total Practical / Practice hours (14 times ‘P’)		
17.	Brief description of module-wise activities pertaining to self-learning component (Only for 700 / 800 level courses) (Include topics that the students would do selflearning from books / resource materials: Do not Include assignments / term papers etc.)		
	Module no.	Description	
18.	Suggested texts and reference materials STYLE: Author name and initials, Title, Edition, Publisher, Year		
	1. Fundamentals of Nuclear Physics by DRS Somayajulu, Jagdish Verma, Roop Chand Bhandari, (2005) CBS Publishers & distributers, New Delhi 2. Introductory Nuclear Physics by R.K. Puri and V.K. Babbar, (1996) Narosa Publishing House 3. Nuclear Physics by Irving Kaplan (1962) Addison-Wesley Publishing Company 4. Nuclear and Particle Physics by Kenneth Kane, Wiley & sons, Singapore (1988) latest edition 5. Introduction to Nuclear and Particle Physics (World Scientific) - A Das, T. Ferbel (2005) (2nd Edition) 6. Particles and Nuclei (Springer) by Povh, Roth, Scholar, Zetsche (2008) Springer-Verlag Berlin Heidelberg, Sixth Edition. 7. Particle Physics a Comprehensive Introduction by Abraham Seiden (2004) (Pearson) 8. Quarks and Leptons (Wiley) by Halzen and Martin, (1984) John Wiley & Sons 9. The Big and the Small- Vol. II: The Facinating Link Between Particle Physics and Cosmology, G Venkataraman		
19.	Resources required for the course (itemized student access requirements, if any)		
	19.1	Software	Name of software, number of licenses, etc.
	19.2	Hardware	Nature of hardware, number of access points, etc
	19.3	Teaching aids (videos, etc.)	Description, Source, etc. Projection System
	19.4	Laboratory	Type of facility required, number of students etc
	19.5	Equipment	Type of equipment required, number of access points, etc.

	19.6	Classroom infrastructure	Type of facility required, number of students etc. Yes
	19.7	Site visits	Type of Industry/ Site, typical number of visits, number of students etc.
		Others (please specify)	
20.	Design content of the course (Percent of student time with examples, if possible)		
	Design-type problems		
	Open-ended problems		
	Project-type activity		
	Open-ended laboratory work		
	Others (please specify)		

COURSE OUTCOMES (COs)

1) **Learn** two body problems in Nuclear Physics.
2) **Analyse** the nuclear forces and the nuclear models
3) **Describe** the nuclear decays and nuclear reaction
4) **Learn** nuclear detectors and accelerators
5) **Classify** symmetries with the Quark model and standard model

Course Articulation Matrix													
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	3	2	3	1	1	1	1	1	1	1	2	1	1
CO2	3	2	3	1	1	1	1	1	1	-	2	3	3
CO3	3	2	3	1	1	1	1	1	1	-	2	2	2
CO4	3	1	3	2	1	1	1	1	1	-	3	2	2
CO5	3	2	3	1	1	1	1	1	1	-	2	3	3

PSUC602 Atomic and Molecular Physics

1.	Department/Centre/School proposing the course		PHYSICS	
2.	Course Title		Atomic and molecular Physics	
3.	L-T-P structure		3-1-0	
4.	Credits		4	
5.	Course number		PS864	
6.	Course Status (Course Category for Program) PG			
	Programme Core for:		M.Sc. Physics	
7.	Pre-requisite(s)		NIL	
8.	Status vis-à-vis other courses			
	8.1 List of courses precluded by taking this course (significant overlap)			
	(a)	Significant Overlap with any UG/PG course of the Dept./Centre/ School	NIL	
	(b)	Significant Overlap with any UG/PG course of other Dept./Centre/ School	NIL	
9.	Not allowed for		NIL	
10.	Frequency of offering		Semester -3	
11.	Faculty who will teach the course: Dr. Shweta Dabhi / Dr. Tapobroto Bhanja			
12.	Will the course require any visiting faculty?		No	
13.	Course contents: One electron atoms (Hydrogen like atoms), Stark effect, Zeeman effect, Paschen-Back effect, Many electron atoms, L-S and J-J coupling, The spectra of the alkalies, Helium and the alkaline earths, Molecular Spectra, NR, Raman spectra			
14.	Lecture Outline(with topics and number of lectures)			
	Module no.	Topic		No. of hours
	1	Review of Atomic systems: Atomic nature of matter, Black body radiation, Photoelectric effect, X-rays and the Compton effect, The Stern-Gerlach experiment – Angular momentum and spin, Space quantization, de Broglie's hypothesis. Fine structure of hydrogenic atoms, The Lamb shift, Hyperfine structure and isotope shifts. The periodic system of the elements, L-S coupling and j-j coupling,		10
	2	The ground state of two-electron systems, para and ortho states, The spectra of the alkali metals, Helium and the alkaline metals		12
	3	Interaction with external electric and magnetic fields: The stark effect and The Zeeman effect, Paschen-Back effect		08
	4	Electronic structure of diatomic molecules, Homonuclear and heteronuclear diatomic molecules, the rotation and vibration of diatomic molecules, The electronic spin and Hund's cases, The structure of polyatomic molecules		12

		Rotational spectra of diatomic molecules, Rayleigh and Raman scattering, vibrational-rotational spectra of diatomic molecules, electronic spectra of diatomic molecules,	
	5	Electron spin, Nuclear magnetic resonance spectroscopy	08
15.	Brief description of tutorial activities:		
	Module no.	Description	No. of hours
	1	Tutorials based on Hydrogen spectra, Zeeman effect, L-S and J-J coupling	5
	2	Rotational Spectra, NMR	5
	Total Tutorial hours (14 times 'T')		
16.	Brief description of Practical / Practice activities		
	Module no.	Description	No. of hours
	Total Practical / Practice hours (14 times 'P')		
17.	Brief description of module-wise activities pertaining to self-learning component (Only for 700 / 800 level courses) (Include topics that the students would do self-learning from books / resource materials: Do not Include assignments / term papers etc.)		
	Module no.	Description	
18.	Suggested texts and reference materials STYLE: Author name and initials, Title, Edition, Publisher, Year		
	Text Book: 1. Physics of Atoms and Molecules by B. H. Bransden and C. J. Joachain, Pearson Publication Reference Book: 1. Atomic and Molecular Spectra: Laser – by Raj Kumar (Kedar Nath Ram Nath publications) 2. Spectroscopy (Atomic and molecular) by G. Chatwal and S. Anand. 3. Elements of spectroscopy – by Gupta, Kumar and Sharma (Pragati Prakashan Meerut)		
19.	Resources required for the course (itemized student access requirements, if any)		
	19.1	Software	Nil
	19.2	Hardware	Nil
	19.3	Teaching aids (videos, etc.)	Description, Source, etc. Projection System
	19.4	Laboratory	Type of facility required, number of students etc
	19.5	Equipment	Type of equipment required, number of access points, etc.
	19.6	Classroom infrastructure	Yes
	19.7	Site visits	Type of Industry/ Site, typical number of visits, number of students etc.
	19.8	Others (please specify)	
20.	Design content of the course (Percent of student time with examples, if possible)		
	Design-type problems		Nil
	Open-ended problems		Nil

	Project-type activity	Nil
	Open-ended laboratory work	Nil
	Others (please specify)	Nil

COURSE OUTCOMES (COs)

1) **Understand** the nature of approximations made on the quantum description of atomic and molecular systems.
2) **Describe** the atomic spectra of one and two valence electron atoms
3) **Understand** the interaction of atoms with external electric and magnetic field
4) **Explain** rotational, vibrational, electronic and Raman spectra of molecules.
5) **Describe** electron spin and nuclear magnetic resonance spectroscopy and their applications.

Course Articulation Matrix													
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	3	2	3	1	1	1	1	1	1	1	2	1	1
CO2	3	2	3	1	1	1	1	1	1	-	2	3	3
CO3	3	2	3	1	1	1	1	1	1	-	2	2	2
CO4	3	1	3	2	1	1	1	1	1	-	3	2	2
CO5	3	2	3	1	1	1	1	1	1	-	2	3	3

PSUC603 Quantum Mechanics – II

1.	Department/Centre/School proposing the course		PHYSIC	
2.	Course Title		Quantum Mechanics – II	
3.	L-T-P structure		3-1-0	
4.	Credits		4	
5.	Course number		PS**	
6.	Course Status (Course Category for Program) PG			
	Programme Core for:		M.Sc. Physics	
7.	Pre-requisite(s)		NIL	
8.	Status vis-à-vis other courses			
	8.1 List of courses precluded by taking this course (significant overlap)			
	(a)	Significant Overlap with any UG/PG course of the Dept./Centre/ School	NIL	
	(b)	Significant Overlap with any UG/PG course of other Dept./Centre/ School	NIL	
9.	Not allowed for		NIL	
10.	Frequency of offering (check one box)		Once in an academic year	
11.	Faculty who will teach the course:			
12.	Will the course require any visiting faculty?		No	
13.	Course objectives “On successful completion of this course, a student should be able to understand the advance concepts of quantum mechanics with applications”			
14.	Course contents: Time dependent Perturbation, Quantum Scattering, Identical Particles, Relativistic Quantum mechanics			
15.	Lecture Outline(with topics and number of lectures)			
	Module no.	Topic		No. of hours
	1	**Identical particles:** Symmetric and anti-symmetric wave functions, Spin ½ systems, Slater determinant, Pauli Exclusion Principle**,** Fermi Gas		5
	2	**Time Dependent Perturbations**: Constant Perturbation, Harmonic Perturbations, Transition probability, Fermi’s Golden Rule, Spontaneous and Stimulated emissions, Dipole Transitions, Selection rules for dipole transitions, Decays, Lifetime and widths, Adiabatic and Sudden Approximations		16
	3	**Quantum Scattering I:** Classical Scattering, Scattering from a potential (General Treatment), Partial wave analysis, Born approximation, Born Series, Ramsauer-Townsend effect, Green’s		15

		function method of scattering, complex potential and absorption, Collisions in composite systems	
	4	**Relativistic Quantum Mechanics**: Klein-Gordon equation, Dirac Matrices and Dirac Equation, Covariant formulation of Dirac equation, Plane wave solutions and central potentials, Non-relativistic limit of Dirac Equation, Concept of particles and antiparticles, Dirac particles in an external electromagnetic field	15
16.	Brief description of tutorial activities:		
	Module no.	Description	No. of hours
	1	Tutorials on time dependent perturbation	2
	2	Tutorials on scattering theory	3
	3	Tutorials on Identical Particles	2
	4	Tutorials on applications: Transitions, Einstein's coefficients, Stark Effect, etc.	2
	Total Tutorial hours (12 times 'T')		
17.	Brief description of Practical / Practice activities		
	Module no.	Description	No. of hours
	Module 1	Solving Scattering problems using scilab/Mathematica	2
	Total Practical / Practice hours (12 times 'P')		
18.	Brief description of module-wise activities pertaining to self-learning component (Include topics that the students would do self-learning from books / resource materials: Do not Include assignments / term papers etc.)		
	Module no.	Description	
	---	---	
19.	Suggested texts and reference materials STYLE: Author name and initials, Title, Edition, Publisher, Year		
	1. Quantum Mechanics Concepts and Applications, N. Zettilli, (Wiley) 2. Introductory Quantum Mechanics, Richard L. Liboff (Pearson). 3. Quantum Mechanics, Bransden Joachain 4. Quantum Mechanics: Walter Greiner 5. Quantum Mechanics: Mathews Venkatesan 6. Quantum Mechanics: Albert Messiah 7. Quantum Mechanics: Cohen Tannoudji		
20.	Resources required for the course (itemized student access requirements, if any)		
	20.1	Software	Scilab/Mathematica
	20.2	Hardware	No hardware required
	20.3	Teaching aids (videos, etc.)	Not as such
	20.4	Laboratory	Computer Laboratory
	20.5	Equipment	Computer
	20.6	Classroom infrastructure	No specific facility required
	20.7	Site visits	Not required
	20.8	Others (please specify)	None

21.	Design content of the course (Percent of student time with examples, if possible)	
	Design-type problems	-
	Open-ended problems	-
	Project-type activity	-
	Open-ended laboratory work	-
	Others (please specify)	-

CO1: At the end of the course, student will be able to understand the basic concepts of time dependent perturbation, scattering, Identical particles, Relativistic quantum mechanics.
CO2: Solve basic and some advanced problems in harmonic perturbations, decay widths and lifetimes, Adiabatic and sudden approximations, quantum scattering (Partial waves and Green's function).
CO3: Student will learn advanced quantum mechanical theory: Relativistic quantum mechanics, concept of Dirac particles and antiparticles, and applications like Stark effect, stimulated and spontaneous emissions, etc.

Course Articulation Matrix											
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	-	-	-	-	3	-	3	-	-	3
CO2	3	-	3	-	-	3	-	-	-	-	3
CO3	3	-	3	-	-	3	-	-	-	-	3

PSUS601 Physics Laboratory- 3

21.	Department/Centre/School proposing the course		PHYSICS
22.	Course Title		Physics Laboratory- 3
23.	L-T-P structure		3-1-0
24.	Credits		4
25.	Course number		
26.	Course Status (Course Category for Program) PG		
	Programme Core for:		M.Sc. Physics
27.	Pre-requisite(s)		NIL
28.	Status vis-à-vis other courses		
	28.1 List of courses precluded by taking this course (significant overlap)		
	(a)	Significant Overlap with any UG/PG course of the Dept./Centre/ School	NIL
	(b)	Significant Overlap with any UG/PG course of other Dept./Centre/ School	NIL
29.	Not allowed for		NIL
30.	Frequency of offering		Semester -3
31.	Faculty who will teach the course: Dr. Manan N. Shah /Dr. Shweta Dabhi / Dr. Tapobroto Bhanja / Dr. C K Sumesh		
32.	Will the course require any visiting faculty?		No
33.	Course contents: One electron atoms (Hydrogen like atoms), Stark effect, Zeeman effect, Paschen-Back effect, Many electron atoms, L-S and J-J coupling, The spectra of the alkalies, Helium and the alkaline earths, Molecular Spectra, NR, Raman spectra		
34.	Lecture Outline(with topics and number of lectures)		

	Module no.	Topic	No. of hours
	1	To determine work function using photo electric effect and verify inverse square law of radiation.	06
	2	To determine the first excitation potential of gas (Argon) by Franck-Hertz experiment	06
	3	i) To measure the wavelengths of visible spectral lines in Balmer series of	06
	4	atomic hydrogen ii) To determine the value of Rydberg's constant	06
	5	Measurement of susceptibility of paramagnetic solution (Quinck`s Tube Method)	06
	6	To measure the resistivity of a semiconductor with temperature by four-probe method (room temperature to 150ºC) and to determine its band gap.	06
	7	To determine Hall coefficient of semiconductor at room temperature.	06

	8	To measure dielectric constant of a ferroelectric material as a function of temperature.	06
	9	Study of the characteristics of a GM tube and determination of its operating voltage and plateau length.	06
	10	To carry out the absorption studies on beta rays with the aid of a GM counter and hence to determine the end point energy of beta rays emitted from a radioactive source.	06
	11	Iodine absorption spectra	06
35.	Brief description of tutorial activities:		
	Module no.	Description	No. of hours
	1	Calculation	10
	2	Graphs	10
	Total Tutorial hours (14 times 'T')		
36.	Brief description of Practical / Practice activities		
	Module no.	Description	No. of hours
	Total Practical / Practice hours (14 times 'P')		
37.	Brief description of module-wise activities pertaining to self-learning component (Only for 700 / 800 level courses) (Include topics that the students would do self-learning from books / resource materials: Do not Include assignments / term papers etc.)		
	Module no.	Description	
38.	Suggested texts and reference materials STYLE: Author name and initials, Title, Edition, Publisher, Year		
	Reference Book: 1. Nuclear Physics by Irving Kaplan (1962) Addison-Wesley Publishing Company 2. Spectroscopy (Atomic and molecular) by G. Chatwal and S. Anand. 3. Elements of spectroscopy – by Gupta, Kumar and Sharma (Pragati Prakashan Meerut)		
39.	Resources required for the course (itemized student access requirements, if any)		
	19.9	Software	Nil
	19.10	Hardware	Nil
	19.11	Teaching aids (videos, etc.)	Description, Source, etc. Projection System
	19.12	Laboratory	Type of facility required, number of students etc
	19.13	Equipment	Type of equipment required, number of access points, etc.
	19.14	Classroom infrastructure	Yes
	19.15	Site visits	Type of Industry/ Site, typical number of visits, number of students etc.
	19.16	Others (please specify)	
40.	Design content of the course (Percent of student time with examples, if possible)		
	Design-type problems	Nil	
	Open-ended problems	Nil	
	Project-type activity	Nil	
	Open-ended laboratory work	Nil	
	Others (please specify)	Nil	

COURSE OUTCOMES (COs)

1) **Understand** the operation/working of the instrument.
2) **Understand** the fundamentals of the experiments which are included in this course and their advanced applications.
3) **Explain** practical knowledge to co-relate with the theoretical studies.

Course Articulation Matrix													
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	3	2	3	1	1	1	1	1	1	1	2	1	1
CO2	3	2	3	1	1	1	1	1	1	-	2	3	3
CO3	3	2	3	1	1	1	1	1	1	-	2	2	2
CO4	3	1	3	2	1	1	1	1	1	-	3	2	2
CO5	3	2	3	1	1	1	1	1	1	-	2	3	3

PSUP601 Minor Project

41.	Department/Centre/School proposing the course		PHYSICS
42.	Course Title		Physics Laboratory- 3
43.	L-T-P structure		14
44.	Credits		7
45.	Course number		
46.	Course Status (Course Category for Program) PG		
	Programme Core for:		M.Sc. Physics
47.	Pre-requisite(s)		NIL
48.	Status vis-à-vis other courses		
	48.1 List of courses precluded by taking this course (significant overlap)		
	(a)	Significant Overlap with any UG/PG course of the Dept./Centre/ School	NIL
	(b)	Significant Overlap with any UG/PG course of other Dept./Centre/ School	NIL
49.	Not allowed for		NIL
50.	Frequency of offering		Semester -3
51.	Faculty who will teach the course: Dr. Manan N. Shah /Dr. Shweta Dabhi / Dr. Tapobroto Bhanja / Dr. C K Sumesh		
52.	Will the course require any visiting faculty?		No
53.	Course contents: One electron atoms (Hydrogen like atoms), Stark effect, Zeeman effect, Paschen-Back effect, Many electron atoms, L-S and J-J coupling, The spectra of the alkalies, Helium and the alkaline earths, Molecular Spectra, NR, Raman spectra		
54.	Lecture Outline(with topics and number of lectures)		

	Module no.	Topic	No. of hours
	11	Iodine absorption spectra	06
55.	Brief description of tutorial activities:		
	Module no.	Description	No. of hours
	1		
	Total Tutorial hours (14 times ‘T’)		
56.	Brief description of Practical / Practice activities		
	Module no.	Description	No. of hours
	Total Practical / Practice hours (14 times ‘P’)		
57.	Brief description of module-wise activities pertaining to self-learning component (Only for 700 / 800 level courses) (Include topics that the students would do self-learning from books / resource materials: Do not Include assignments / term papers etc.)		

	Module no.		Description
58.	Suggested texts and reference materials STYLE: Author name and initials, Title, Edition, Publisher, Year		
59.	Resources required for the course (itemized student access requirements, if any)		
	19.17	Software	Nil
	19.18	Hardware	Nil
	19.19	Teaching aids (videos, etc.)	Description, Source, etc. Projection System
	19.20	Laboratory	Type of facility required, number of students etc
	19.21	Equipment	Type of equipment required, number of access points, etc.
	19.22	Classroom infrastructure	Yes
	19.23	Site visits	Type of Industry/ Site, typical number of visits, number of students etc.
	19.24	Others (please specify)	
60.	Design content of the course (Percent of student time with examples, if possible)		
	Design-type problems		Nil
	Open-ended problems		Nil
	Project-type activity		Nil
	Open-ended laboratory work		Nil
	Others (please specify)		Nil

COURSE OUTCOMES (COs)

CO1: **Formulate** research problem based on literature survey.

CO2: **Identify** and **demonstrate** appropriate research methodology (such as an experiment, theory or simulation) that will answer the research question.

CO3: **Implement** problem solving skills to constructively address research setbacks

Course Articulation Matrix

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	-	3	-	-	-	-	-	-	-	3
CO2	3	-	3	3	-	-	-	3	-	-	3
CO3	-	3	3	3	-	-	-	-	-	-	3

Semester 4

PSUP602 Research Project

61.	Department/Centre/School proposing the course		PHYSICS
62.	Course Title		Research Project
63.	L-T-P structure		0-0-36
64.	Credits		23
65.	Course number		
66.	Course Status (Course Category for Program) PG		
	Programme Core for:		M.Sc. Physics
67.	Pre-requisite(s)		NIL
68.	Status vis-à-vis other courses		
	68.1 List of courses precluded by taking this course (significant overlap)		
	(a)	Significant Overlap with any UG/PG course of the Dept./Centre/ School	NIL
	(b)	Significant Overlap with any UG/PG course of other Dept./Centre/ School	NIL
69.	Not allowed for		NIL
70.	Frequency of offering		Semester -4
71.	Faculty who will teach the course: Dr. Manan N. Shah /Dr. Shweta Dabhi / Dr. Tapobroto Bhanja / Dr. C K Sumesh		
72.	Will the course require any visiting faculty?		No
73.	Course contents: One electron atoms (Hydrogen like atoms), Stark effect, Zeeman effect, Paschen-Back effect, Many electron atoms, L-S and J-J coupling, The spectra of the alkalies, Helium and the alkaline earths, Molecular Spectra, NR, Raman spectra		
74.	Lecture Outline(with topics and number of lectures)		

	Module no.	Topic	No. of hours
75.	Brief description of tutorial activities:		
	Module no.	Description	No. of hours
	1		
	Total Tutorial hours (14 times ‘T’)		
76.	Brief description of Practical / Practice activities		
	Module no.	Description	No. of hours
	Total Practical / Practice hours (14 times ‘P’)		
77.	Brief description of module-wise activities pertaining to self-learning component (Only for 700 / 800 level courses) (Include topics that the students would do self-learning from books / resource materials: Do not Include assignments / term papers etc.)		

	Module no.		Description
78.	Suggested texts and reference materials STYLE: Author name and initials, Title, Edition, Publisher, Year		
79.	Resources required for the course (itemized student access requirements, if any)		
	19.25	Software	Nil
	19.26	Hardware	Nil
	19.27	Teaching aids (videos, etc.)	Description, Source, etc. Projection System
	19.28	Laboratory	Type of facility required, number of students etc
	19.29	Equipment	Type of equipment required, number of access points, etc.
	19.30	Classroom infrastructure	Yes
	19.31	Site visits	Type of Industry/ Site, typical number of visits, number of students etc.
	19.32	Others (please specify)	
80.	Design content of the course (Percent of student time with examples, if possible)		
	Design-type problems		Nil
	Open-ended problems		Nil
	Project-type activity		Nil
	Open-ended laboratory work		Nil
	Others (please specify)		Nil

COURSE OUTCOMES (COs)

At the end of the research project , student would be able to

CO1: **Carry** out a substantial research-based project

CO2: **Learn** to analyse the data/output and synthesize research findings through computational understanding or analytical skills required for the experimentalist.

CO3: **Write** a complete project report (dissertation).

CO4: **Deliver** a clear and effective presentation and develop skill for communication and research discussion which will be useful for doctoral research proposal presentation.

Course Articulation Matrix

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	-	3	-	-	3	3	-	-	3	-	3
CO2	-	-	3	3	-	-	-	-	-	-	3
CO3	-	3	-	3	-	-	-	-	-	-	3
CO4	-	3	-	3	-	3	-	3	-	-	3

DEPARTMENTAL ELECTIVES

PSE1 Thin film Technology

1.	Department/Centre/School proposing the course		PHYSICS	
2.	Course Title		Thin film Technology	
3.	L-T-P structure		3-0-1	
4.	Credits		4	
5.	Course number		PSE1	
6.	Course Status (Course Category for Program) PG			
	Programme Core for:		M.Sc. Physics	
7.	Pre-requisite(s)		NIL	
8.	Status vis-à-vis other courses			
	8.1 List of courses precluded by taking this course (significant overlap)			
	(a)	Significant Overlap with any UG/PG course of the Dept./Centre/ School	NIL	
	(b)	Significant Overlap with any UG/PG course of other Dept./Centre/ School	NIL	
9.	Not allowed for		NIL	
10.	Frequency of offering		Elective in any semester	
11.	Faculty who will teach the course:		Dr. C. K. Sumesh, Dr. Vanraj Solanki	
12.	Will the course require any visiting faculty?		No	
13.	Course contents: Growth and types of thin films, Physical vapor deposition, Chemical Vapor deposition, Solution based deposition, Characterization of thin films, Applications of thin films			
14.	Lecture Outline(with topics and number of lectures)			
	Module no.	Topic		No. of hours
	1	Basics of thin films, nucleation and growth, capillarity theory, atomistic and kinetic models of nucleation, stages of film growth and mechanisms, structure, epitaxy, types of thin films, metal, semiconductor and insulator		10
	2	Different types of Physical vapor deposition techniques and Chemical Vapor deposition techniques		5
	3	Electrochemical deposition, dip-coating, spin-coating, chemical bath deposition, direct liquid coating, spray pyrolysis, sol gel technique, ink-jet material printing		10
	4	Characterization of thin films: X-ray diffraction, Scanning Electron Microscopy, Atomic Force Microscopy, optical properties		10
	5	Applications of thin films: Photodetectors, protective coatings, solar cells, optical coatings, decorative coatings, batteries		10

15.	Brief description of tutorial activities:		
	Module no.	Description	No. of hours
	Total Tutorial hours (14 times 'T')		
16.	Brief description of Practical / Practice activities		
	Module no.	Description	No. of hours
	Total Practical / Practice hours (14 times 'P')		
17.	Brief description of module-wise activities pertaining to self-learning component		
	Module no.	Description	
18.	Suggested texts and reference materials STYLE: Author name and initials, Title, Edition, Publisher, Year		
	1. Thin Film Fundamentals by A. Goswami 2. Handbook of Deposition Technologies for Films and Coatings by P. M. Martin (Editor) 3. Solution Processing of Inorganic Materials Edited by David Mitzi 4. Materials Science of Thin Films by M. Ohring		
19.	Resources required for the course (itemized student access requirements, if any)		
	19.1	Software	Nil
	19.2	Hardware	Nil
	19.3	Teaching aids (videos, etc.)	Description, Source, etc. Projection System
	19.4	Laboratory	Type of facility required, number of students etc
	19.5	Equipment	Type of equipment required, number of access points, etc.
	19.6	Classroom infrastructure	Yes
	19.7	Site visits	Type of Industry/ Site, typical number of visits, number of students etc.
	19.8	Others (please specify)	
20.	Design content of the course (Percent of student time with examples, if possible)		
	Design-type problems		
	Open-ended problems		
	Project-type activity		
	Open-ended laboratory work		
	Others (please specify)		

Course Outcomes:

CO1: Gain an understanding of thin film structure and characteristics.

CO2: Gain a working understanding of the link between deposition methods, film structure, and film attributes.

CO3: The student will gain experience with a range of thin film applications.

Course Articulation Matrix

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	-	-	-	-	3	-	-	-	-	3
CO2	-	-	3	3	-	3	-	3	3	-	3
CO3	3	-	3	-	-	3	-	-	-	-	3

PSE2 Physics of Solar cells

1.	Department/Centre/School proposing the course		PHYSICS	
2.	Course Title		Physics of Solar cells	
3.	L-T-P structure		3-0-1	
4.	Credits		4	
5.	Course number		PSE2	
6.	Course Status (Course Category for Program) PG			
	Programme Core for:		M.Sc. Physics	
7.	Pre-requisite(s)		NIL	
8.	Status vis-à-vis other courses			
	8.1 List of courses precluded by taking this course (significant overlap)			
	(a)	Significant Overlap with any UG/PG course of the Dept./Centre/ School	NIL	
	(b)	Significant Overlap with any UG/PG course of other Dept./Centre/ School	NIL	
9.	Not allowed for		NIL	
10.	Frequency of offering		Elective in any semester	
11.	Faculty who will teach the course:		Dr. C. K. Sumesh, Dr. Vanraj Solanki	
12.	Will the course require any visiting faculty?		No	
13.	Course contents: Importance of Solar Energy, P-N Junction as solar cells, Design of solar cell, Monocrystalline silicon (Si) solar cell, Thin film solar cells, Emerging solar cells			
14.	Lecture Outline(with topics and number of lectures)			
	Module no.	Topic		No. of hours
	1	Energy scenario, Need for sustainable energy, Renewable Energy sources, Role of Solar Energy, Sun as energy resource, Solar Radiation: Extra terrestrial and terrestrial radiations, instruments to measure solar radiation, solar radiation geometry, hourly, daily and seasonal variation of insolation,solar radiation on horizontal and tilted surfaces		10
	2	P-N Junction as solar cells Space charge region, Energy band diagram, P-N junction potential and depletion region, Carrier Movements and concentration profiles, P-N junction in Non-equilibrium condition, P-N junction under illumination, photovoltaic effect, generation and recombination of carriers, light generated current, I-V equations and characteristics of solar cell		10
	3	Design of solar cell Upper limit of solar cell parameters, Losses in solar cells, Design for High Isc, Design for High Voc, Design for FF, I-V		10

		and QE measurements, Minority carrier life time and diffusion length measurements	
	4	Monocrystalline silicon (Si) solar cell Production of Metallurgical and electronics Grade Si, Production of Monocrystalline Si, Processes for fabrication of Monocrystalline Si, High efficiency Si solar cells	10
	5	Thin film solar cells Advantage of Thin Film technology, Materials for Thin film Technology, Features of Thin film Technology, Thin film Deposition techniques, Cadmium Telluride (CdTe) solar cells, Chalcopyrite (CIS/CIGS) solar cells, Polycrystalline Si solar cells	7
	6	Emerging solar cells: Organic/polymer solar cells, Dye-sensitized solar cells (DSC), Quantum dot solar cells (QDSCs), CZTS solar cells	5
15.	Brief description of tutorial activities:		
	Module no.	Description	No. of hours
	Total Tutorial hours (14 times 'T')		
16.	Brief description of Practical / Practice activities		
	Module no.	Description	No. of hours
17.	Brief description of module-wise activities pertaining to self-learning component (Only for 700 / 800 level courses) (Include topics that the students would do self-learning from books / resource materials: Do not Include assignments / term papers etc.)		
	Module no.	Description	
18.	Suggested texts and reference materials STYLE: Author name and initials, Title, Edition, Publisher, Year		
	1. The Physics of Solar Cells by Jenny Nelson (Imperial College Press, 2013) 2. Physics of Solar Cells by Peter Wurfel 3. Solar Photovoltaics: Fundamentals, Technologies and Applications by Chetan Solanki 4. Thin film Solar Cells by K. L. Chopra and S. R. Das		
19.	Resources required for the course (itemized student access requirements, if any)		
	19.1	Software	Nil
	19.2	Hardware	Nil
	19.3	Teaching aids (videos, etc.)	Description, Source, etc. Projection System
	19.4	Laboratory	Type of facility required, number of students etc
	19.5	Equipment	Type of equipment required, number of access points, etc.
	19.6	Classroom infrastructure	Yes
	19.7	Site visits	Type of Industry/ Site, typical number of visits, number of students etc.
	19.8	Others (please specify)	

20.	Design content of the course (Percent of student time with examples, if possible)	
	Design-type problems	
	Open-ended problems	
	Project-type activity	
	Open-ended laboratory work	
	Others (please specify)	

Course Outcomes:

CO1: The students would learn and understand various photovoltaics technologies

CO2: The students would be able to understand Solar cell fundamentals and operation of a solar cell

CO3: The students would be able to understand design of silicon solar cells

Course Articulation Matrix											
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	-	-	-	-	-	-	-	-	-	3
CO2	3	3	-	-	-	-	-	-	-	3	3
CO3	3	3	3	3	-	-	-	-	3	3	3

PSE3 Nanomaterials: Synthesis, Properties and Applications

<table>
<tr><td>1.</td><td colspan="2">Department/Centre/School proposing the course</td><td colspan="3">PHYSICS</td></tr>
<tr><td>2.</td><td colspan="2">Course Title</td><td colspan="3">Nanomaterials: Synthesis, Properties and Applications</td></tr>
<tr><td>3.</td><td colspan="2">L-T-P structure</td><td colspan="3">3-0-1</td></tr>
<tr><td>4.</td><td colspan="2">Credits</td><td>4</td><td>on-graded Units</td><td>Please fill appropriate details in S. No. 21</td></tr>
<tr><td>5.</td><td colspan="2">Course number</td><td colspan="3">PSE3</td></tr>
<tr><td>6.</td><td colspan="5">Course Status (Course Category for Program) PG</td></tr>
<tr><td></td><td colspan="2">Programme Core for:</td><td colspan="3">M.Sc. Physics</td></tr>
<tr><td>7.</td><td colspan="2">Pre-requisite(s)</td><td colspan="3">NIL</td></tr>
<tr><td>8.</td><td colspan="5">Status vis-à-vis other courses</td></tr>
<tr><td></td><td colspan="5">8.1 List of courses precluded by taking this course (significant overlap)</td></tr>
<tr><td></td><td>(a)</td><td>Significant Overlap with any UG/PG course of the Dept./Centre/ School</td><td colspan="3">NIL</td></tr>
<tr><td></td><td>(b)</td><td>Significant Overlap with any UG/PG course of other Dept./Centre/ School</td><td colspan="3">NIL</td></tr>
<tr><td>9.</td><td colspan="2">Not allowed for</td><td colspan="3">NIL</td></tr>
<tr><td>10.</td><td colspan="2">Frequency of offering</td><td colspan="3">Elective in any semester</td></tr>
<tr><td>11.</td><td colspan="2">Faculty who will teach the course:</td><td colspan="3">Dr. Kinnari Parekh/Dr. C K Sumesh</td></tr>
<tr><td>12.</td><td colspan="2">Will the course require any visiting faculty?</td><td colspan="3">No</td></tr>
<tr><td>13.</td><td colspan="5">Course contents:
Nanotechnology – development history, Overview on characterization and synthesis of nanostructure materials, Overview of nanostructures, Nanodevices, New fields of nanotechnology, Implications of nanotechnology, Synthesis of Nanomaterials (Physical Methods, Chemical Methods, Biological Methods), Properties of Nanomaterials</td></tr>
<tr><td>14.</td><td colspan="5">Lecture Outline(with topics and number of lectures)</td></tr>
<tr><td></td><td>Module no.</td><td colspan="3">Topic</td><td>No. of hours</td></tr>
<tr><td></td><td>1</td><td colspan="3">NANOSCALE SYSTEMS: Length scales in physics, Nanostructures: 1D, 2D and 3D nanostructures (nanodots, thin films, nanowires, nanorods), Band structure and density of states of materials at nanoscale, Size Effects in nano systems, Quantum confinement: Applications of Schrodinger equation- Infinite potential well, potential step, potential box, quantum confinement of carriers in 3D, 2D, 1D nanostructures and its consequences.</td><td>10</td></tr>
<tr><td></td><td>2</td><td colspan="3">SYNTHESIS OF NANOSTRUCTURE MATERIALS: Top down and Bottom up approach, Photolithography. Ball milling. Gas phase condensation. Vacuum deposition. Physical vapor deposition</td><td>10</td></tr>
</table>

		(PVD): Thermal evaporation, E-beam evaporation, Pulsed Laser deposition. Chemical vapor deposition (CVD). Sol-Gel. Electro deposition. Spray pyrolysis. Hydrothermal synthesis. Preparation through colloidal methods. MBE growth of quantum dots.	
	3	CHARACTERIZATION: X-Ray Diffraction. Optical Microscopy. Scanning Electron Microscopy. Transmission Electron Microscopy. Atomic Force Microscopy. Scanning Tunneling Microscopy.	5
	4	OPTICAL PROPERTIES: Coulomb interaction in nanostructures. Concept of dielectric constant for nanostructures and charging of nanostructure. Quasi- articles and excitons. Excitons in direct and indirect band gap semiconductor nanocrystals. Quantitative treatment of quasi-particles and excitons, charging effects. Radiative processes: General formalization-absorption, emission and luminescence. Optical properties of heterostrctures and nanostructures.	10
	5	ELECTRON TRANSPORT: Carrier transport in nanostrcutures. Coulomb blockade effect, thermionic emission, tunneling and hoping conductivity. Defects and impurities: Deep level and surface defects.	5
	6	APPLICATIONS: Applications of nanoparticles, quantum dots, nanowires and thin films for photonic devices (LED, solar cells). Single electron devices (no derivation). CNT based transistors. Nanomaterial Devices: Quantum dots heterostructure lasers, optical switching and optical data storage. Magnetic quantum well; magnetic dots - magnetic data storage. Micro Electromechanical Systems (MEMS), Nano Electromechanical Systems (NEMS).	10
15.	Brief description of tutorial activities:		
	Module no.	Description	No. of hours
	Total Tutorial hours (14 times ‘T’)		
16.	Brief description of Practical / Practice activities		
	Module no.	Description	No. of hours
	Total Practical / Practice hours (14 times ‘P’)		
17.	Brief description of module-wise activities pertaining to self-learning component (Only for 700 / 800 level courses) (Include topics that the students would do self-learning from books / resource materials: Do not Include assignments / term papers etc.)		
	Module no.	Description	
18.	Suggested texts and reference materials STYLE: Author name and initials, Title, Edition, Publisher, Year		
	Text Books: Essentials of nanotechnology by Jeremy Ramsden [JR], 2009, Jeremy Ramsden & Ventus Publishing ApS, (download from http://bookboon.com/en/nano-technology-ebook) Nanotechnology: Principles and practices, Sulabha K Kulkarni, Capital publishing company, 2007.		

	Reference Books: 1. Introduction to Nanoscience, S.M.Lindsay, Oxford ISBN 978-019-954421-9 (2010). 2. Guozhong Cao (2004). Nanostructures and Nanomaterials: Synthesis, Properties & Applications, 448 pages, Imperial College Press, ISBN-10:1860944159 3. Introduction to Nanoscience, S.M.Lindsay, Oxford ISBN 978-019-954421-9 (2010). 4. Nanocrystalline Materials, A.I. Gusev, A.A. Rempel, Cambridge International Science Publishing, (2003) • C.P. Poole, Jr. Frank J. Owens, Introduction to Nanotechnology (Wiley India Pvt. Ltd.). • S.K. Kulkarni, Nanotechnology: Principles & Practices (Capital Publishing Company) • K.K. Chattopadhyay and A. N. Banerjee, Introduction to Nanoscience and Technology (PHI Learning Private Limited). • Richard Booker, Earl Boysen, Nanotechnology (John Wiley and Sons). • M. Hosokawa, K. Nogi, M. Naita, T. Yokoyama, Nanoparticle Technology Handbook (Elsevier, 2007). • Bharat Bhushan, Springer Handbook of Nanotechnology (Springer-Verlag, Berlin, 2004).		
19.	Resources required for the course (itemized student access requirements, if any)		
	19.1	Software	Nil
	19.2	Hardware	Nil
	19.3	Teaching aids (videos, etc.)	Description, Source, etc. Projection System
	19.4	Laboratory	Type of facility required, number of students etc
	19.5	Equipment	Type of equipment required, number of access points, etc.
	19.6	Classroom infrastructure	Yes
	19.7	Site visits	Type of Industry/ Site, typical number of visits, number of students etc.
	19.8	Others (please specify)	
20.	Design content of the course (Percent of student time with examples, if possible)		
	Design-type problems		
	Open-ended problems		

Course Outcomes:

CO1: The student will gain a basic understanding of nanomaterials.

CO2: Recognize and comprehend various top-down and bottom-up nanomaterial synthesis methodologies.

CO3: The student will gain experience in utilizing nanomaterials' unique features to a variety of applications.

Course Articulation Matrix											
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	-	-	-	-	3	-	-	-	-	3
CO2	-	-	3	3	-	3	-	3	3	-	3
CO3	3	-	3	-	-	3	-	-	-	-	3

PSE4 Semiconductor Devices

1.	Department/Centre/School proposing the course		PHYSICS
2.	Course Title		Semiconductor Devices
3.	L-T-P structure		4-0-0
4.	Credits		4
5.	Course number		PSE4
6.	Course Status (Course Category for Program) PG		
	Programme Core for:		M.Sc. Physics
7.	Pre-requisite(s)		NIL
8.	Status vis-à-vis other courses		
	8.1 List of courses precluded by taking this course (significant overlap)		
	(a)	Significant Overlap with any UG/PG course of the Dept./Centre/ School	NIL
	(b)	Significant Overlap with any UG/PG course of other Dept./Centre/ School	NIL
9.	Not allowed for		NIL
10.	Frequency of offering		Elective in any semester
11.	Faculty who will teach the course:		Dr. C. K. Sumesh,
12.	Will the course require any visiting faculty?		No
13.	Course contents: METAL-SEMICONDUCTOR CONTACTS, METAL-INSULATOR-SEMICONDUCTOR CAPACITORS, HETEROJUNCTIONS, HETEROJUNCTION BIPOLAR TRANSISTOR, MOSFETs, MESFETs, and MODFETs		
14.	Lecture Outline(with topics and number of lectures)		

	Module no.	Topic	No. of hours
	1	METAL-SEMICONDUCTOR CONTACTS: Introduction, formation of barrier, Ideal Condition, Depletion Layer, Interface States, Image-Force Lowering, Barrier-Height Adjustment, Current transport processes, Measurement of barrier height, device structures, Ohmic contact.	15
	2	METAL-INSULATOR-SEMICONDUCTOR CAPACITORS: Introduction, Ideal Mis Capacitor, Surface Space-Charge Region, Ideal MIS Capacitance Curves, Silicon Mos Capacitor, Interface Traps, Carrier Transport, Accumulation- and Inversion-Layer Thickness	15
	3	HETEROJUNCTIONS, HETEROJUNCTION BIPOLAR TRANSISTOR: Anisotype Heterojunction, Isotype Heterojunction, Heterojunction bipolar transistor, Double-Heterojunction	15

		Bipolar Transistor, Graded-Base Bipolar Transistor, Hot-Electron Transistor	
	4	MOSFETs, MESFETs, and MODFETs: Introduction, Basic Device Characteristics, Nonuniform Doping and Buried-Channel Device, Device Scaling And Short-Channel Effects, MOSFET Structures, Circuit Applications, Nonvolatile Memory Devices, Single-Electron Transistor, MESFET, MODFET,I-V Characteristics, Device Structures	15
15.	Brief description of tutorial activities:		
	Module no.	Description	No. of hours
	1		
	2		
	Total Tutorial hours (14 times 'T')		
16.	Brief description of Practical / Practice activities		
	Module no.	Description	No. of hours
	Total Practical / Practice hours (14 times 'P')		
17.	Brief description of module-wise activities pertaining to self-learning component (Only for 700 / 800 level courses) (Include topics that the students would do self-learning from books / resource materials: Do not Include assignments / term papers etc.)		
	Module no.	Description	
18.	Suggested texts and reference materials STYLE: Author name and initials, Title, Edition, Publisher, Year		
	1. Thin Film Fundamentals by A. Goswami 2. Handbook of Deposition Technologies for Films and Coatings by P. M. Martin (Editor) 3. Solution Processing of Inorganic Materials Edited by David Mitzi 4. Materials Science of Thin Films by M. Ohring		
19.	Resources required for the course (itemized student access requirements, if any)		
	19.1	Software	Nil
	19.2	Hardware	Nil
	19.3	Teaching aids (videos, etc.)	Description, Source, etc. Projection System
	19.4	Laboratory	Type of facility required, number of students etc
	19.5	Equipment	Type of equipment required, number of access points, etc.
	19.6	Classroom infrastructure	Yes
	19.7	Site visits	Type of Industry/ Site, typical number of visits, number of students etc.
	19.8	Others (please specify)	

20.	Design content of the course (Percent of student time with examples, if possible)	
	Design-type problems	
	Open-ended problems	
	Project-type activity	
	Open-ended laboratory work	
	Others (please specify)	

Course Outcomes:

CO1: Explain material properties and semiconductor electronics applications.

CO2: Demonstrate the operation of basic electronic devices using semiconductor expertise.

CO3: Identify and classify semiconductor devices that are used in specific applications.

Course Articulation Matrix

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	-	-	-	-	3	-	-	-	-	3
CO2	-	-	3	3	-	3	-	3	3	-	3
CO3	3	-	3	-	-	3	-	-	-	-	3

PSE5 Experimental techniques

1.	Department/Centre/School proposing the course		PHYSICS	
2.	Course Title		Experimental techniques	
3.	L-T-P structure		3-0-1	
4.	Credits		4	
5.	Course number		PSE5	
6.	Course Status (Course Category for Program) PG			
	Programme Core for:		M.Sc. Physics	
7.	Pre-requisite(s)		NIL	
8.	Status vis-à-vis other courses			
	8.1 List of courses precluded by taking this course (significant overlap)			
	(a)	Significant Overlap with any UG/PG course of the Dept./Centre/ School	NIL	
	(b)	Significant Overlap with any UG/PG course of other Dept./Centre/ School	NIL	
9.	Not allowed for		NIL	
10.	Frequency of offering		Elective in any semester	
11.	Faculty who will teach the course:		Dr. C. K. Sumesh,	
12.	Will the course require any visiting faculty?		No	
13.	Course contents: Error and uncertainty analysis in measurements, Signals and Systems, Shielding and Grounding, Transducers & industrial instrumentation (working principle, efficiency, applications), Linear Position transducer, Strain gauge, Piezoelectric. Inductance change transducer, Digital Multimeter, Vacuum Systems			
14.	Lecture Outline(with topics and number of lectures)			
	Module no.	Topic		No. of hours
	1	Measurements: Accuracy and precision. Significant figures. Error and uncertainty analysis. Types of errors: Gross error, systematic error, random error. Statistical analysis of data (Arithmetic mean, deviation from mean, average deviation, standard deviation, chi-square) and curve fitting. Guassian distribution.		5
	2	Signals and Systems: Periodic and aperiodic signals. Impulse response, transfer function and frequency response of first and second order systems. Fluctuations and Noise in measurement system. S/N ratio and Noise figure. Noise in frequency domain. Sources of Noise: Inherent fluctuations, Thermal noise, Shot noise, 1/f noise		10
	3	Shielding and Grounding: Methods of safety grounding. Energy coupling. Grounding. Shielding: Electrostatic shielding. Electromagnetic Interference.		5

	4	Transducers & industrial instrumentation (working principle, efficiency, applications): Static and dynamic characteristics of measurement Systems. Generalized performance of systems, Zero order first order, second order and higher order systems. Electrical, Thermal and Mechanical systems. Calibration. Transducers and sensors. Characteristics of Transducers. Transducers as electrical element and their signal conditioning. Temperature transducers: RTD, Thermistor, Thermocouples, Semiconductor type temperature sensors (AD590, LM35, LM75) and signal conditioning. Linear Position transducer	10
	5	Strain gauge, Piezoelectric. Inductance change transducer: Linear variable differential transformer (LVDT), Capacitance change transducers. Radiation Sensors: Principle of Gas filled detector, ionization chamber, scintillation detector.	5
	6	Digital Multimeter: Comparison of analog and digital instruments. Block diagram of digital multimeter, principle of measurement of I, V, C. Accuracy and resolution of measurement. Impedance Bridges and Q-meter: Block diagram and working principles of RLC bridge. Q-meter and its working operation. Digital LCR bridge.	10
	7	Vacuum Systems: Characteristics of vacuum: Gas law, Mean free path. Application of vacuum. Vacuum system- Chamber, Mechanical pumps, Diffusion pump & Turbo Modular pump, Pumping speed, Pressure gauges (Pirani, Penning, ionization).	10
15.	Brief description of tutorial activities:		
	Module no.	Description	No. of hours
	1		
	2		
	Total Tutorial hours (14 times ‘T’)		
16.	Brief description of Practical / Practice activities		
	Module no.	Description	No. of hours
	Total Practical / Practice hours (14 times ‘P’)		
17.	Brief description of module-wise activities pertaining to self-learning component		
	Module no.	Description	
18.	Suggested texts and reference materials STYLE: Author name and initials, Title, Edition, Publisher, Year		
	• Measurement, Instrumentation and Experiment Design in Physics and Engineering, M. Sayer and A. Mansingh, PHI Learning Pvt. Ltd.		

	• Experimental Methods for Engineers, J.P. Holman, McGraw Hill • Introduction to Measurements and Instrumentation, A.K. Ghosh, 3rd Edition, PHI Learning Pvt. Ltd. • Transducers and Instrumentation, D.V.S. Murty, 2nd Edition, PHI Learning Pvt. Ltd. • Instrumentation Devices and Systems, C.S. Rangan, G.R. Sarma, V.S.V. Mani, Tata McGraw Hill • Principles of Electronic Instrumentation, D. Patranabis, PHI Learning Pvt. Ltd. • Electronic circuits: Handbook of design & applications, U.Tietze, Ch.Schenk, Springer		
19.	Resources required for the course (itemized student access requirements, if any)		
	19.1	Software	Nil
	19.2	Hardware	Nil
	19.3	Teaching aids (videos, etc.)	Description, Source, etc. Projection System
	19.4	Laboratory	Type of facility required, number of students etc
	19.5	Equipment	Type of equipment required, number of access points, etc.
	19.6	Classroom infrastructure	Yes
	19.7	Site visits	Type of Industry/ Site, typical number of visits, number of students etc.
	19.8	Others (please specify)	
20.	Design content of the course (Percent of student time with examples, if possible)		
	Design-type problems		
	Open-ended problems		
	Project-type activity		
	Open-ended laboratory work		
	Others (please specify)		

PSE6 Plasma Physics

1.	Department/Centre/School proposing the course		PHYSICS	
2.	Course Title		Plasma Physics	
3.	L-T-P structure		3-0-1	
4.	Credits		4	
5.	Course number		PSE9	
6.	Course Status (Course Category for Program) PG			
	Programme Core for:		M.Sc. Physics	
7.	Pre-requisite(s)		NIL	
8.	Status vis-à-vis other courses			
	8.1 List of courses precluded by taking this course (significant overlap)			
	(a)	Significant Overlap with any UG/PG course of the Dept./Centre/ School	NIL	
	(b)	Significant Overlap with any UG/PG course of other Dept./Centre/ School	NIL	
9.	Not allowed for		NIL	
10.	Frequency of offering		Elective in any semester	
11.	Faculty who will teach the course:		Dr. C. K. Sumesh	
12.	Will the course require any visiting faculty?		No	
13.	Course contents: PLASMA PHYSICS FUNDAMENTALS, LABORATORY PLASMA PRODUCTION, PLASMA DIAGNOSTIC TECHNIQUES, INTRODUCTION TO FUSION ENERGY			
14.	Lecture Outline(with topics and number of lectures)			
	Module no.	Topic		No. of hours
	1	PLASMA PHYSICS FUNDAMENTALS: Definition of Plasma, Concept of Temperature, Debye Shielding, Single particle motions, Plasmas as fluids, Fluid drift in magnetic field, Waves in Plasmas, Group Velocity, Elementary Plasma Waves, Single Fluid MHD equation, Diffusion in fully ionized plasma, Bohm diffusion, Hydromagnetic equilibrium, Two-stream instabillity, "Gravitational" instability, Equations of Kinetic Theory, Landau Damping, Ion-Acoustic Shock wave, Parametric instabilities.		14
	2	LABORATORY PLASMA PRODUCTION: Assembled Plasmas, Irradiated Gases, Electrical Discharges, Self-ionized gases, Optical Radiation.		9
	3	PLASMA DIAGNOSTIC TECHNIQUES: Electric and Magnetic Probes, Spectral Intensities, UV, Far-Infrared and X-ray spectroscopy, Optical interferometry, Microwave Techniques.		10

	4	INTRODUCTION TO FUSION ENERGY: Basic Principles, Plasma confinement and Transport, Plasma Heating and Fueling, Advanced confinement concepts, Fusion engineering, Fusion Reactor Technology.	10
15.	Brief description of tutorial activities:		
	Module no.	Description	No. of hours
	1		
	2		
	Total Tutorial hours (14 times 'T')		
16.	Brief description of Practical / Practice activities		
	Module no.	Description	No. of hours
	Total Practical / Practice hours (14 times 'P')		
17.	Brief description of module-wise activities pertaining to self-learning component		
	Module no.	Description	
	1	Production of a Glow Discharge	
	2	Production of a Hot Cathode Discharge and Measurement Of Plasma Density And Temperature Using Langmuir Probes.	
	3	Extraction of Ions from Plasma	
	4	Cusp Confinement Plasma System	
18.	Suggested texts and reference materials STYLE: Author name and initials, Title, Edition, Publisher, Year		
	• Introduction to Plasma Physics, by Francis F Chen, Plenum Press New York 1977. • Plasma Diagnostic Technique, by R H Huddlestone and S L Leonard, Academic Press 1965. • Plasma Physics in Theory and Application, by W B Kunkel, McGrawHill, New York 1966. • Introduction to Fusion Energy, by J Reece roth, Ibis Publishing, Virginia, 1986.		
19.	Resources required for the course (itemized student access requirements, if any)		
	19.1	Software	Nil
	19.2	Hardware	Nil
	19.3	Teaching aids (videos, etc.)	Description, Source, etc. Projection System
	19.4	Laboratory	Type of facility required, number of students etc
	19.5	Equipment	Type of equipment required, number of access points, etc.
	19.6	Classroom infrastructure	Yes
	19.7	Site visits	Type of Industry/ Site, typical number of visits, number of students etc.
	19.8	Others (please specify)	
20.	Design content of the course (Percent of student time with examples, if possible)		

	Design-type problems	
	Open-ended problems	
	Project-type activity	
	Open-ended laboratory work	
	Others (please specify)	

Course Outcomes:

At the end of the course, the student will be able to

CO1: solve simple problems charged particle in motion, magnetised fluid systems in equilibrium and wave propagation in plasmas

CO2: formulate the conditions for a plasma to be in a state of thermodynamic equilibrium, or non-equilibrium, and analyse the stability of this equilibrium

CO3: understand the complex physical phenomena observed in plasma.

Course Articulation Matrix											
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	-	3	-	-	3	-	-	-	-	3
CO2	3	-	3	3	-	3	-	3	3	-	3
CO3	3	-	3	-	-	3	-	-	-	-	3

PSE7 General Relativity, Black holes and Cosmology – I			
1.	Department/Centre/School proposing the course		PHYSICS
2.	Course Title		General Relativity, Black holes and Cosmology - I
3.	L-T-P structure		4
4.	Credits		4
5.	Course number		PSE7
6.	Course Status (Course Category for Program) PG		
	Programme Core for:		M.Sc. Physics
7.	Pre-requisite(s)		Basic linear algebra, vectors and tensors, basic knowledge on Special Theory of Relativity
8.	Status vis-à-vis other courses		
	8.1 List of courses precluded by taking this course (significant overlap)		
	(a)	Significant Overlap with any UG/PG course of the Dept./Centre/ School	NIL
	(b)	Significant Overlap with any UG/PG course of other Dept./Centre/ School	NIL
9.	Not allowed for		NIL
10.	Course offered in		Semester 2
11.	Faculty who will teach the course: Dr. Dipanjan Dey		
12.	Will the course require any visiting faculty?		No
13.	Course contents: Detail mathematical and conceptual understanding of General Relativity, Black holes and Cosmology.		
14.	Lecture Outline(with topics and number of lectures)		
	Module no.	Topic	No. of hours
	1	**Vectors & Tensors:** vectors, Tensors, basis vectors, contravariant and covariant vectors and tensors, transformation properties of contravariant and covariant tensors, tensor calculus in curved manifold, covariant and contravariant derivative, Gauss law and Stokes law in higher dimension, Christoffel symbols and its properties.	10
	2	**Inertial, Non-inertial frames & Equivalence Principle:** Principles of special relativity, spacetime continuum, Lorentz transformation, Minkowski spacetime, 4-vectors, Non-inertial frame, equivalence principle, space-time structure of an accelerated frame with respect to an inertial frame, Difficulties with Euclidean geometry, rotating dish, curvilinear coordinate system, Riemannian Geometry.	10

	3	**Geodesics & Conserved Quantities:** Parallel transport and covariant derivative, coordinate basis and ortho-normal basis, importance of ortho-normal basis, local flatness and mathematical interpretation of equivalence principle, Parallel transport of 4-velocity and geodesic equation, geodesic equation from least action principle and conserved quantities, affine Geodesic and metric geodesic, Lie derivative.	10
	4	**Riemann Curvature Tensor and Einstein Field Equations:** Geodesic deviation, Riemann curvature tensor and its properties, Newtonian limit of Riemann curvature tensor, Energy-momentum tensor, Einstein field equation, solution of Einstein field equation, energy conditions.	08
	5	**Solutions of Einstein Equations and Blackholes:** Spacetime around a star, Collapse of a star, Schwarzschild spacetime, geodesic properties of Schwarzschild spacetime, perihelion precession, causal structures (with different causal diagrams) of Schwarzschild spacetime and Schwarzschild Black-hole, rotating blackholes and Kerr metric.	12
	6	**Expanding Universe and Cosmology**: Newtonian Cosmology and Hubble's law, Relativistic Cosmology: Derivation of FLRW metric from cosmological principle, Matter in Cosmology: energy momentum tensor for relativistic and non-relativistic matter, scalar field, Freidman equations and its solutions for different matter components, cosmological horizons and Hubble scale, Cosmological Redshift.	10

16	Brief description of tutorial activities:		
	Module no.	Description	No. of hours
		Total Tutorial hours	
17	Brief description of Practical / Practice activities		
	Module no.	Description	No. of hours
18	Suggested texts and reference materials :		

1. A first course in General Relativity, Bernard Schutz.
2. An introduction to Einstein General relativity, James B Hartle
3. General Relativity: An introduction for Physicist, M.P. Hobson
4. General Relativity, Robert Wald
5. Mathematical Theory of Black Holes, S Chandrasekhar
6. Physical Foundations of Cosmology, Mukhanov

19	Brief description of module-wise activities pertaining to self-learning component (Include topics that the students would do self-learning from books / resource materials: Do not Include assignments / term papers etc.)	
	Module no.	Description

20	Resources required for the course (itemized student access requirements, if any)		
	19.1	Software	Nil
	19.2	Hardware	Nil
	19.3	Teaching aids (videos, etc.)	Writing board, Projection System
	19.4	Laboratory	Nil
	19.5	Equipment	Nil
	19.6	Classroom infrastructure	Yes
	19.7	Site visits	Nil
	19.8	Others (please specify)	
21	Design content of the course (Percent of student time with examples, if possible)		
	Design-type problems		20% of student time of practice hours
	Open-ended problems		Nil
	Project-type activity		Nil
	Open-ended laboratory work		Nil
	Others (please specify)		Nil

Course Outcomes:

CO1: Basic concepts of General theory of relativity and basic understanding of the importance of General relativity in modern Astrophysics and Cosmology.

CO2: Students will be able to solve conceptual and technical problems on General theory relativity and Cosmology.

CO3: Students will get an introduction of general relativistic astrophysics and cosmology which will encourage them to explore deep into the field in future.

Course Articulation Matrix											
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	3	3	3	-	-	-	3	-	-	3
CO2	3	3	3	3	-	-	-	3	-	-	3
CO3	3	3	3	3	-	-	-	3	-	-	3

<table>
<tr><th colspan="5">PSE8 General Relativity, Black holes and Cosmology – II</th></tr>
<tr><td>1.</td><td colspan="2">Department/Centre/School proposing the course</td><td colspan="2">PHYSICS</td></tr>
<tr><td>2.</td><td colspan="2">Course Title</td><td colspan="2">General Relativity, Black holes and Cosmology - II</td></tr>
<tr><td>3.</td><td colspan="2">L-T-P structure</td><td colspan="2">4</td></tr>
<tr><td>4.</td><td colspan="2">Credits</td><td colspan="2">4</td></tr>
<tr><td>5.</td><td colspan="2">Course number</td><td colspan="2">PSE8</td></tr>
<tr><td>6.</td><td colspan="4">Course Status (Course Category for Program) PG</td></tr>
<tr><td></td><td colspan="2">Programme Core for:</td><td colspan="2">M.Sc. Physics</td></tr>
<tr><td>7.</td><td colspan="2">Pre-requisite(s)</td><td colspan="2">Basic linear algebra, vectors and tensors, basic knowledge on Special Theory of Relativity and General Relativity, Black holes and Cosmology - I</td></tr>
<tr><td>8.</td><td colspan="4">Status vis-à-vis other courses</td></tr>
<tr><td></td><td colspan="4">8.1 List of courses precluded by taking this course (significant overlap)</td></tr>
<tr><td></td><td>(a)</td><td>Significant Overlap with any UG/PG course of the Dept./Centre/ School</td><td colspan="2">NIL</td></tr>
<tr><td></td><td>(b)</td><td>Significant Overlap with any UG/PG course of other Dept./Centre/ School</td><td colspan="2">NIL</td></tr>
<tr><td>9.</td><td colspan="2">Not allowed for</td><td colspan="2">NIL</td></tr>
<tr><td>10.</td><td colspan="2">Course offered in</td><td colspan="2">Semester 3</td></tr>
<tr><td>11.</td><td colspan="4">Faculty who will teach the course: Dr. Dipanjan Dey</td></tr>
<tr><td>12.</td><td colspan="2">Will the course require any visiting faculty?</td><td colspan="2">No</td></tr>
<tr><td>13.</td><td colspan="4">Course contents:
Detail mathematical and conceptual understanding of General Relativity, Black holes and Cosmology.</td></tr>
<tr><td>14.</td><td colspan="4">Lecture Outline(with topics and number of lectures)</td></tr>
<tr><td></td><td colspan="2">Module no.</td><td>Topic</td><td>No. of hours</td></tr>
<tr><td></td><td colspan="2">1</td><td>Geodesic Congruences and Hypersurfaces: Kinematics of deformable medium: expansion, shear, rotation, Congruence of timelike and null geodesics: Transverse metric, Frobenius theorem, Raychaudhuri’s equation, Focusing theorem, interpretation of expansion scalar, description of hypersurfaces, Integration on hypersurfaces, Intrinsic and extrinsic curvature, Gauss-Codazzi equations, initial value problems, Junction conditions.</td><td>20</td></tr>
<tr><td></td><td colspan="2">2</td><td>Lagrangian and Hamiltonian formalism:. Lagrangian formulation and Hamiltonian formulation of General Relativity.</td><td>10</td></tr>
<tr><td></td><td colspan="2">3</td><td>Black Holes: Schwarzschild Black holes (revision), Reissner-Nordstrom black hole and its causal structures, Kerr Black holes and its causal structures.</td><td>15</td></tr>
</table>

	4	**Topological spaces, differential forms and General Relativity**	10
	5	**Cosmological perturbations:** Linear cosmological perturbations to FLRW metric and energy momentum tensor (hydrodynamic fluid and scalar field), adiabatic and entropy perturbations, perturbed Einstein equations and perturbed continuity equations, scalar vector tensor decompositions, introduction to different gauges and gauge invariant quantities, Perturbation equation in different gauges and in terms of gauge invariant quantities, different applications of cosmological perturbations (brief over-view).	15
	6	**Expanding Universe and Cosmology**: Newtonian Cosmology and Hubble's law, Relativistic Cosmology: Derivation of FLRW metric from cosmological principle, Matter in Cosmology: energy momentum tensor for relativistic and non-relativistic matter, scalar field, Freidman equations and its solutions for different matter components, cosmological horizons and Hubble scale, Cosmological Redshift.	10

16	Brief description of tutorial activities:		
	Module no.	Description	No. of hours
	Total Tutorial hours		
17	Brief description of Practical / Practice activities		
	Module no.	Description	No. of hours

18	Suggested texts and reference materials :
	1.A first course in General Relativity, Bernard Schutz. 2. An introduction to Einstein General relativity, James B Hartle 3. General Relativity: An introduction for Physicist, M.P. Hobson 4. General Relativity, Robert Wald 5. Mathematical Theory of Black Holes, S Chandrasekhar 6. Large Scale Structure of Spacetime, Hawking and Ellis 7. Physical Foundations of Cosmology, Mukhanov

19	Brief description of module-wise activities pertaining to self-learning component (Include topics that the students would do self-learning from books / resource materials: Do not Include assignments / term papers etc.)	
	Module no.	Description
	-----	-----

20	Resources required for the course (itemized student access requirements, if any)		
	20.1	Software	Nil
	20.2	Hardware	Nil

	20.3	Teaching aids (videos, etc.)	Writing board, Projection System
	20.4	Laboratory	Nil
	20.5	Equipment	Nil
	20.6	Classroom infrastructure	Yes
	20.7	Site visits	Nil
	20.8	Others (please specify)	
21	Design content of the course (Percent of student time with examples, if possible)		
	Design-type problems		20% of student time of practice hours
	Open-ended problems		Nil
	Project-type activity		Nil
	Open-ended laboratory work		Nil
	Others (please specify)		Nil

Course Outcomes:

CO1: Introductory concepts of modern General relativistic Astrophysics and cosmology.

CO2: Students will be able to solve conceptual and difficult technical problems on General theory relativity and Cosmology.

CO3: Students will be familiar with the different research problems on Astrophysics and Cosmology and can start doing research on the field of Astrophysics and Cosmology.

Course Articulation Matrix

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	3	3	3	-	-	-	3	-	-	3
CO2	3	3	3	3	-	-	-	3	-	-	3
CO3	3	3	3	3	-	-	-	3	-	-	3

PSE9 Vacuum Technology

1.	Department/Centre/School proposing the course		PHYSICS	
2.	Course Title		Vacuum Technology	
3.	L-T-P structure		3-0-1	
4.	Credits		4	
5.	Course number		PSE9	
6.	Course Status (Course Category for Program) PG			
	Programme Core for:		M.Sc. Physics	
7.	Pre-requisite(s)		NIL	
8.	Status vis-à-vis other courses			
	8.1 List of courses precluded by taking this course (significant overlap)			
	(a)	Significant Overlap with any UG/PG course of the Dept./Centre/ School	NIL	
	(b)	Significant Overlap with any UG/PG course of other Dept./Centre/ School	NIL	
9.	Not allowed for		NIL	
10.	Frequency of offering		Elective in any semester	
11.	Faculty who will teach the course:		Dr. Manan N. Shah, Dr. C. K. Sumesh,	
12.	Will the course require any visiting faculty?		No	
13.	Course contents: PHYSICS OF RAREFIED GASES, PRODUCTION OF VACUUM, MEASUREMENT OF VACUUM PARAMETERS, VACUUM SYSTEMS, COMPONENTS, APPLICATIONS OF VACUUM TECHNOLOGY			
14.	Lecture Outline(with topics and number of lectures)			
	Module no.	Topic		No. of hours
	1	PHYSICS OF RAREFIED GASES: Kinetic Theory, Gas Laws, Mean Free path, Transport Processes, Viscosity of gases, Diffusion of gases, Resistance and conductance of vacuum pipework, Gas flow in pipework, Pump-down Time, Adsorption and Desorption of gases, Solubility of gases, Permeation of gases through solids, Vapour pressure, Thermal dissociation of Metallic Oxides		12
	2	PRODUCTION OF VACUUM: Classification, types, and ranges of vacuum pumps, Vacuum pump variables, Baffles and Traps.		8
	3	MEASUREMENT OF VACUUM PARAMETERS: Classification, types and ranges of vacuum gauges, Measurement of Partial pressure, Flow measurement, Leak Testing.		8
	4	VACUUM SYSTEMS, COMPONENTS: Vacuum System engineering, Vacuum joints, Seals, Valves, feed-through, View ports, Vacuum materials.		10
	5	APPLICATIONS OF VACUUM TECHNOLOGY: Vacuum systems and components for Electronics, Metallurgy, Chemical and Nuclear fields.		7

15.	Brief description of tutorial activities:		
	Module no.	Description	No. of hours
	1		
	2		
	Total Tutorial hours (14 times 'T')		
16.	Brief description of Practical / Practice activities		
	Module no.	Description	No. of hours
	Total Practical / Practice hours (14 times 'P')		
17.	Brief description of module-wise activities pertaining to self-learning component		
	Module no.	Description	
	1	Evacuation of a System Using a Water Ring Pump & Use of Dial Gauge	
	2	Evacuation of a System Using Rotary Vacuum Pump & Use of Pirani Gauge	
	3	Evacuation of a System Using a Diffusion Pump & Use Of Penning Gauge	
	4	Measurement of Helium Leak Rate Using MSLD Technique	
18.	Suggested texts and reference materials STYLE: Author name and initials, Title, Edition, Publisher, Year		
	• Fundamentals of Vacuum Techniques, by A Pipko, V. Pliskovsky, B. Korolev, V. Kuznetsov, Mir Publishers, Moscow 1984. • Scientific Foundations of Vacuum Technique, by S Dushman, John Wiley & Sons Inc. New York 1962. • Vacuum Technology and Applications, by D J Hucknall, Butterworth-Heinemann, Oxford, 1991.		
19.	Resources required for the course (itemized student access requirements, if any)		
	19.9	Software	Nil
	19.10	Hardware	Nil
	19.11	Teaching aids (videos, etc.)	Description, Source, etc. Projection System
	19.12	Laboratory	Type of facility required, number of students etc
	19.13	Equipment	Type of equipment required, number of access points, etc.
	19.14	Classroom infrastructure	Yes
	19.15	Site visits	Type of Industry/ Site, typical number of visits, number of students etc.
	19.16	Others (please specify)	
20.	Design content of the course (Percent of student time with examples, if possible)		
	Design-type problems		
	Open-ended problems		
	Project-type activity		
	Open-ended laboratory work		
	Others (please specify)		

Course Outcomes:

Student will be able to
CO1: The students will gain insight into the various measurement parameters of detection of vacuum.
CO2: The student will learn the methodological concepts and tools needed to acquire top-level skills in the field of vacuum technology.
CO3: The student will gain the sound knowledge of vacuum related technology in the context of applied research or industrial R&D.

Course Articulation Matrix											
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	-	3	-	-	-	-	-	-	-	3
CO2	3	-	3	3	-	-	-	3	3	-	3
CO3	3	-	3	-	-	-	-	-	3	-	3

CS291 Data Analytic

1.	Department/Centre/School proposing the course	DEPSTAR	
2.	Course Title	Data Analytic	
3.	L-T-P structure	4-0-0	
4.	Credits	4	
5.	Course number	CS291	
6.	Course Status (Course Category for Program) PG		
	Programme Core for:	M.Sc. Physics	
7.	Pre-requisite(s)	NIL	
8.	Status vis-à-vis other courses		
	8.1 List of courses precluded by taking this course (significant overlap)		
	(a)	Significant Overlap with any UG/PG course of the Dept./Centre/ School	NIL
	(b)	Significant Overlap with any UG/PG course of other Dept./Centre/ School	NIL
9.	Not allowed for	NIL	
10.	Frequency of offering (check one box)	Elective in any semester	
11.	Faculty who will teach the course:		
12.	Will the course require any visiting faculty?	No	
13.	Course contents: Fundamental concepts of Data Science, Statistical Learning Techniques, Data Pre-processing, Supervised Learning, Unsupervised Learning, Reinforcement learning, Evaluation		
14.	Lecture Outline(with topics and number of lectures)		
	Module no.	Topic	No. of hours
	1	Introduction to Data Science, significance of Data Science in today's digitally-driven world, applications of Data Science, lifecycle of Data Science, issues in Data Science	4
	2	Descriptive statistics, Measures of Central Tendency, Simple Linear Regression, ANOVA, Logistic Regression, Multi Linear regression, Correlation, Moving Average, Random Number Generation, Histogram, Sampling, Percentile Rank, t-test, chi-square, z-test	14
	3	Why to pre-process data?, Data cleaning: Missing Values, Noisy Data, Data Integration and transformation, Data Reduction: Data cube aggregation, Dimensionality reduction, Data Compression, Numerosity Reduction, Data Visulization	10
	4	Classification, Decision Trees, Random Forest Classifier, Naïve Bayes Classifier, Support Vector Machine, K - Nearest Neighbours, Artificial Neural Networks	12
	5	Clustering, K-means, Hierarchical clustering, Density based clustering, Association Rules, Dimensionality Reduction - Principal Component Analysis	12
	6	Q Learning, Non deterministic rewards and Actions	04

	7	Cross-Validation, Measures of Performance for Classification (Accuracy, Confusion Matrix, Precision, Recall, F1-Score), Measures of Performance for Clustering (Homogeneity, Completeness, V-Measure)	04
15.	Brief description of tutorial activities:		
	Module no.	Description	No. of hours
	1		
	2		
	Total Tutorial hours (14 times ‘T’)		
16.	Brief description of Practical / Practice activities		
	Module no.	Description	No. of hours
	Total Practical / Practice hours (14 times ‘P’)		
17.	Brief description of module-wise activities pertaining to self-learning component (Only for 700 / 800 level courses) (Include topics that the students would do self-learning from books / resource materials: Do not Include assignments / term papers etc.)		
	Module no.	Description	
18.	Suggested texts and reference materials STYLE: Author name and initials, Title, Edition, Publisher, Year		
	Text Books: a. “Machine Learning”, Tom Mitchell, McGraw Hill, 1997. ISBN 0070428077 b. Ethem Alpaydin, “Introduction to Machine Learning”, MIT Press, 2004 c. Hastie, Trevor, et al., “The elements of statistical learning,” Vol. 2. No. 1. New York springer, 2009. d. Montgomery, Douglas C., and George C. Runger. “Applied statistics and probability for engineers,” John Wiley & Sons, 2010. Reference Books: 1. Christopher M. Bishop, “Pattern Recognition and Machine Learning”, Springer, 2006. 2. Richard O. Duda, Peter E. Hart & David G. Stork, “Pattern Classification. Second Edition”, Wiley & Sons, 2001. 3. Trevor Hastie, Robert Tibshirani and Jerome Friedman, “The elements of statistical learning”, Springer, 2001. 4. Richard S. Sutton and Andrew G. Barto, “Reinforcement learning: An introduction”, MIT Press, 1998. Web Materials: 1. https://nptel.ac.in/courses/110106072/ 2. https://www.youtube.com/playlist?list=PLRueFtKLr0QN7MmQ8pdpQerOe_s8vGJG4 3. https://www.youtube.com/watch?v=PPLop4L2eGk&list=PLLssT5z_DsK-h9vYZkQkYNWcItqhlRJLN		
19.	Resources required for the course (itemized student access requirements, if any)		

	19.1	Software	Orange, Weka, Sipina, Tanagra, Easy NN, Statistica, Mathematica
	19.2	Hardware	Nil
	19.3	Teaching aids (videos, etc.)	Description, Source, etc. Projection System
	19.4	Laboratory	Type of facility required, number of students etc
	19.5	Equipment	Type of equipment required, number of access points, etc.
	19.6	Classroom infrastructure	Yes
	19.7	Site visits	Type of Industry/ Site, typical number of visits, number of students etc.
	19.8	Others (please specify)	
20.	Design content of the course (Percent of student time with examples, if possible)		
	Design-type problems		
	Open-ended problems		
	Project-type activity		
	Open-ended laboratory work		
	Others (please specify)		

Course Outcomes:

CO1: Students will be able to use concepts and methods of mathematical disciplines relevant to data analytics and statistical modelling.
CO2: Students will demonstrate proficiency with statistical analysis of Physics data.
CO3: Student will know how to use Machine learning concepts for prediction in Data Analytics.

Course Articulation Matrix

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	1	1	3	3	-	3	-	1	1	-	2
CO2	-	-	3	3	1	2	1	1	1	1	-
CO3	-	-	3	3	-	3	1	2	2	-	2

Semester –III

PSUP601 Minor Project

(Credits: 06)
The total project hours during the semester will be approximately 100 hours.

Semester –IV

PS UP602 Research Project

(Credits: 25)
The total project hours during the semester will be approximately 460 hours.

Objectives:
The main objectives of the Research Project are to demonstrate the following abilities:

- Research the literature and gather background information
- Analyse the methodology (such as an experiment or simulation) that answered the research question in the specified research article
- Develop scientific writing and oral presentation skills

Learning outcomes:

- The literature survey and the methodology development during Research Project-I would provide the background information necessary for the investigations during the research phase of the project in 3^{rd} and 4^{th} semester
- Would be able to design an appropriate methodology (such as an experiment or simulation) that will answer the research question.
- Would be able to provide justification for the approach adopted considering possible sources of bias or error in the methods used.

Project Evaluation

The evaluation of the project shall be done in continues evaluation mode keeping the following criterion through presentation, seminar, viva-voce etc.

- **Introduction and Background**: an introduction of the research paper, clearly stating the hypothesis or objective of the work and motivation for the work. Background to the research work and similar work through exposition of relevant literature. This should contain sufficient information to allow the audience to appreciate the literature survey you have made.
- **Methods employed to carry out the research work (such as an experiment, simulation, theory etc.)**
- **Discussion of the results of the specified research work:** Results and their critical analysis should be mentioned, whether the results conform to expectations or otherwise and how they compare with other related work. Where appropriate evaluation of the work against the original objectives should be presented. Any problems or difficulties and the suggested solutions should be mentioned. Alternative solutions and their evaluation should also be included.
- **Conclusion**: concluding remarks and observations, unsolved problems, suggestions for further work.

The different components of evaluation and the weights assigned to these components are depicted are mentioned in the Appendix I-III

Guidelines for doing project

- The project work provides the opportunity to study a topic in depth that has been chosen or which has been suggested by the teacher.
- PS757 Research Project-1 is to understand the research methodology to equip the students to carry out research projects of 2nd Year.
- The students first carryout a literature survey for at least ten hours and as their area of interest and shall gather the background information. They have to select one recent research paper related to the choice and fix it for further analysis.
- The periodic progress of the project work as per the criterion mentioned above shall be monitored after every 30% of the total project duration.
- The student is required to give a seminar on the project work done. The Project Evaluation Committee (Internal as well as External examiners) would conduct the viva-voce based on each criterion mentioned above.
- The student is required to submit project record book to maintain the 'Day-to-day work'. All concepts, drawings, formulas, derivations, experimental, observations, graphs, charts, photographs, computer flow charts and pseudo codes must be recorded by the student on this notebook, which must be produced before all evaluation committees' boards. There shall no blank pages in between the writings.
- The student is required to submit a written report/Dissertation at the end of the semester.

Appendix-I

Semester –III

PSUP601 Minor Project

(Credits: 06)

The total project hours during the semester will be approximately 100 hours.

Sem III	Subject	Subject	Teaching scheme				Theory Evaluation			Practical Evaluation			Total I +II
			L + T	P	Contact hrs	Credits	Institute	Univ.	Total -I	Institute	Univ.	Total-II	
	PS	Minor Project	-	8	8	6	-	-	-	100	100	200	200

Examination: Internal Assessment

Criterion	Marks	Remarks
Day to Day work	50	Evaluated by supervisors
Mid-Term Presentation	50	Evaluated by Project Evaluation Committee

Total	**100**	

Examination: External Assessment (University Level)

Criterion	Marks	Remarks
Written Project work	**50**	Evaluated by Project Evaluation Committee
Report Presentation and Viva-Voce	**50**	Evaluated by Project Evaluation Committee
Total	**200**	

Course Outcome:

Student would be able to

CO1: Formulate research problem based on literature survey.

CO2: Identify and demonstrate appropriate research methodology (such as an experiment, theory or simulation) that will answer the research question.

CO3: know and apply problem solving skills to constructively address research setbacks

Course Articulation Matrix

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	-	3	-	-	-	-	-	-	-	3
CO2	3	-	3	3	-	-	-	3	-	-	3
CO3	-	3	3	3	-	-	-	-	-	-	3

Appendix-II

Semester –IV

PSUP602 Research Project

(Credits: 25)
The total project hours during the semester will be approximately 460 hours.

Sem IV	Subject	Subject	Teaching scheme				Theory Evaluation			Practical Evaluation			Total I +II
			L+T	P	Contact hrs.	Credits	Institute	Univ.	Total-I	Institute	Univ.	Total-II	
	PS	Research Project-III	-	30	30	25	-	-	-	400	400	800	800

Examination:

Criterion	Marks	Credit	Remarks	
1st Internal Presentation and Viva-Voce	100	3	Evaluated by Project Evaluation Committee (I)	Internal Examination
Day to Day work	150	4	Evaluated by supervisors (I)	Internal Examination
Project based Theory paper	50	2	Evaluated by supervisors (I)	Internal Examination
2nd Internal Presentation and Viva-Voce	100	3	Evaluated by Project Evaluation Committee (I)	Internal Examination
Poster Presentation	50	2	Evaluated by Project Evaluation Committee (E)	University Examination
Report Presentation and Viva-Voce	200	6	Evaluated by Project Evaluation Committee (E)	University Examination
Dissertation Thesis	150	5	Evaluated by Project Evaluation Committee (E)	University Examination
Total	800	25		

Course Outcomes:

At the end of the research project – III, student would be able to
CO1: Carry out a substantial research-based project
CO2: Learn to analyse the data/output and synthesize research findings through computational understanding or analytical skills required for the experimentalist.
CO3: Write a complete project report (dissertation).

CO4: Deliver a clear and effective presentation and develop skill for communication and research discussion which will be useful for doctoral research proposal presentation.

Course Articulation Matrix

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	-	3	-	-	3	3	-	-	3	-	3
CO2	-	-	3	3	-	-	-	-	-	-	3
CO3	-	3	-	3	-	-	-	-	-	-	3
CO4	-	3	-	3	-	3	-	3	-	-	3

Appendix-IV

GUIDELINES FOR USE OF PROJECT RECORD BOOK

1. The Project Record Book constitutes the bonafide record of project work carried out by undergraduate, postgraduate and research students of Department of Physical Sciences, PDPIAS, CHARUSAT.
2. The book contains day to day record of all conceptual, analytical, laboratory and computational activities carried out by a student as a part of his project.
3. It is a permanent record of academic activity and contains intellectual property created by the student and his supervisor.
4. The book should be treated with respect and maintained with care. Pages must not be torn or used for rough work.
5. The student should record all his thoughts, observations, flow charts, computational steps etc., directly on this notebook. Use of second rough book and final copying to this record book is discouraged. No cognizance of those extra books will be taken for evaluation.
6. All information recorded here must start with a date on the left margin. The work of the day must be organized into sections such as objective, experimental or computational methods, observations, program flow charts, pseudo-codes, conclusion, discussion etc., as relevant to the problem at hand. Short computer prints, photographs, charts and graphs may be pasted neatly wherever necessary.
7. The supervisor should examine the progress of the student and record his observations, comments and suggestions in a regular manner, typically once every week.
8. The student must produce this record book before all Project Evaluation Committee for evaluation and grading of his day to day performance, and for award of medals and prizes. The first evaluation of the project by the Supervisor will be made basing on the record book only.
9. On completion of the project, the student must surrender this book to his supervisor for archiving. If the same problem is continued by a student of the following batch, the supervisor may choose to give it to those students for the sake of continuity. Projects with supervisor intellectual material may be sent to Departmental Library for permanent archival.
10. The students who do work worth publishing and/or patenting are advised to proceed with those activities. The IPR Cell of the Institute will organize the patenting process.